

# FIXED NUMERATORS IN THE FORGÁCS–TRAN FAMILY: BOUNDED DEFECT AND ANGULAR ZERO DISCREPANCY

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**ABSTRACT.** Forgács and Tran proved that the polynomial sequence generated by the reciprocal of  $Q(t) + zt^r$ , with  $Q$  having only positive real zeros, is eventually real-rooted with all zeros in an explicit positive interval. Their Problem 14 asks what survives when a proper numerator, equivalently arbitrary fixed polynomial initial conditions, is introduced. No bound can be uniform over all numerators, but for each fixed numerator the number of zeros off the positive ray is bounded independently of the coefficient index, and eventually the same is true outside the Forgács–Tran interval. Restricting the numerator to the denominator curve produces a canonical univariate weight that carries its asymptotic effect. Eventually, both the defect and the uniform discrepancy in the Forgács–Tran angular parameter are bounded linearly in the weight’s degree, yielding a numerator-independent limiting zero distribution. On bulk subarcs away from its zeros the weight gives the numerator-dependent clock correction, while on uniquely dominant branches its zeros attract finite packets of coefficient zeros.

## 1. INTRODUCTION

Forgács and Tran studied polynomial sequences generated by rational functions with denominator  $Q(t) + zt^r$ , where  $Q$  has only positive real zeros. Let  $Q \in \mathbb{R}[t]$  satisfy

$$(1.1) \quad Q(0) > 0, \quad Q(t) = Q(0) \prod_{j=1}^n \left(1 - \frac{t}{x_j}\right), \quad 0 < x_1 \leq \cdots \leq x_n,$$

where multiplicities are allowed, and let  $r \geq 1$ . For the denominator-only sequence

$$(1.2) \quad \frac{1}{Q(t) + zt^r} = \sum_{m \geq 0} H_m(z) t^m,$$

Forgács and Tran proved that every sufficiently large  $H_m$  is real-rooted and that all of its zeros lie in an explicit interval  $I_{Q,r} = (a, b) \subset (0, \infty)$  [12, Thm. 1]. Their proof constructs a strictly increasing parameterization  $z = z(\theta)$ ,  $0 < \theta < \pi/r$ , for which the two minimum-modulus denominator zeros form the conjugate pair

$$t_{\pm}(\theta) = \tau(\theta) e^{\pm i\theta}.$$

We call this pair *principal*. Its interference controls  $H_m(z(\theta))$  and produces the positive zero bulk.

Problem 14 of their earlier paper asks what survives when the numerator is allowed to vary [11]. Fix a real bivariate polynomial  $N(t, z)$  with  $\deg_t N < \deg_t(Q(t) + zt^r)$  and define

$$(1.3) \quad \frac{N(t, z)}{Q(t) + zt^r} = \sum_{m \geq 0} P_m(z) t^m.$$

The numerator is equivalently a fixed vector of polynomial initial conditions for the denominator recurrence; Proposition 2.1 makes this precise. The natural fixed-numerator question is whether the number of zeros of  $P_m$  outside  $(0, \infty)$  remains bounded as  $m \rightarrow \infty$ .

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*Date:* August 23, 2026.

*2020 Mathematics Subject Classification.* Primary 26C10; Secondary 05A15, 30C15.

*Key words and phrases.* real-rooted polynomial, rational generating function, exceptional zero, angular zero discrepancy, zero distribution, clock spacing, residue amplitude, initial conditions.

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This work is archived at doi:10.5281/zenodo.21517117.

No such bound can be uniform over all numerators. If  $R(z) \in \mathbb{R}[z]$  and  $N(t, z) = R(z)$ , then  $P_m(z) = R(z)H_m(z)$ , so every zero of  $R$  occurs in every coefficient polynomial. Thus the uniform-in-numerator reading of Problem 14 has a negative answer. The fixed-numerator problem has a positive one.

### 1.1. Main theorem.

**Theorem 1.1** (Fixed-numerator Forgács–Tran theorem). *Let  $Q \in \mathbb{R}[t]$  be nonconstant and satisfy (1.1). Let  $r \geq 1$  and assume  $\max\{\deg Q, r\} > 1$ . Let  $N \in \mathbb{R}[t, z] \setminus \{0\}$  satisfy  $\deg_t N < \deg_t(Q(t) + zt^r)$ , where  $\deg_t$  is the degree in  $t$  over  $\mathbb{R}[z]$ , so that  $\deg_t(Q(t) + zt^r) = \max\{\deg Q, r\}$ , and define  $P_m \in \mathbb{R}[z]$  by (1.3).*

- (i) *There is a constant  $C = C(Q, r, N)$  such that every nonzero  $P_m$  has at most  $C$  zeros outside  $(0, \infty)$ , counted with multiplicity.*
- (ii) *If  $I_{Q,r} = (a, b)$  is the Forgács–Tran interval defined in (3.1), then every sufficiently large  $P_m$  is nonzero, has at most  $C$  zeros outside  $I_{Q,r}$ , counted with multiplicity, and has at least  $\deg P_m - C$  distinct zeros in  $I_{Q,r}$ . If  $\lambda_N$  is the Laurent valuation in Lemma 2.2, then eventually*

$$\deg P_m = \left\lfloor \frac{m - \lambda_N}{r} \right\rfloor,$$

*and hence  $\deg P_m = m/r + O_{Q,r,N}(1)$ .*

- (iii) *Write the canonical Laurent restriction as*

$$N\left(t, -\frac{Q(t)}{t^r}\right) = t^{\lambda_N} B_N(t), \quad B_N(0) \neq 0.$$

*There are constants  $C_0 = C_0(Q, r)$  and  $C_1 = C_1(Q, r)$ , independent of  $N$ , such that for all sufficiently large  $m$  the interval conclusions in (ii) hold with defect at most  $C_0 + C_1 \deg B_N$ .*

Conclusion (i) holds at every index, while (iii) is eventual. The onset in (iii) can depend on the numerator; whether a comparable linear bound holds at every index is left open in Question 6.1.

**1.2. Proof strategy.** Two reductions expose the numerator dependence. First, the denominator curve  $Q(t) + zt^r = 0$  has coordinate ring  $\mathbb{R}[t, t^{-1}]$ . Restricting the numerator to this curve gives canonically

$$N\left(t, -\frac{Q(t)}{t^r}\right) = t^{\lambda_N} B_N(t), \quad B_N(0) \neq 0,$$

and Lemma 2.2 shows that, after an explicitly bounded transient,  $P_m$  is exactly a shifted coefficient of  $B_N(t)/(Q(t) + zt^r)$ . Thus  $\lambda_N$  is an index shift, while the tail zero geometry is carried by one fixed univariate weight  $B_N$ .

Second, a contour-separated pole expansion retains only denominator roots whose moduli can compete and absorbs all other poles into one exponentially small remainder. On the Forgács–Tran interval the principal conjugate pair is co-dominant, and the normalized coefficient takes the oscillatory form

$$2|W(\theta)| \cos((M+1)\theta - \psi(\theta)) + R_M(\theta),$$

where  $M = m - \lambda_N$ ,  $W$  is the principal residue amplitude, and  $\psi = \arg W$ . Bounded phase variation and endpoint separation give the uniform discrepancy in the Forgács–Tran angular parameter  $\theta$ , from which Theorem 1.1 follows. This is the conjugate-pair extraction and phase-counting strategy used in [1, 21]. Here the phase budget is bounded uniformly over weights of a given degree, so the eventual defect bound of Theorem 1.1 depends on the numerator only through  $\deg B_N$ . Dividing the same discrepancy by the degree of  $P_m$  gives a limiting zero distribution independent of the numerator. On compact subarcs where  $W \neq 0$ , the same expansion becomes exponentially accurate and yields local strong-clock spacing. On a uniquely dominant branch, a zero of the corresponding residue amplitude instead attracts a finite packet of coefficient zeros. The full synthesis of these two spectral regimes appears in Section 6.

**1.3. Prior work.** Beraha–Kahane–Weiss theory identifies limiting zero loci from competition between dominant branches and from zeros of dominant residue amplitudes [2, 19]; Brown and Otto recently extended this limit-set theory beyond the simple-characteristic-root setting [6]. For rational generating functions, Tran developed this viewpoint in several families [20, 21]. A circulating manuscript of Boyer and Tran treats a general numerator for  $D(t, z) = A(z)t^n + B(z)t + 1$  and obtains an algebraic limiting curve together with a finite exceptional set controlled by a resultant [4, Thm. 1]. These are asymptotic descriptions of limit sets. By contrast, Theorem 1.1 bounds the exceptional zeros uniformly in the coefficient index and, eventually, bounds the interval discrepancy linearly in the degree of the canonical reduced weight.

Related bounded-defect phenomena occur for other recurrences and pencils. Mai bounded the number of zeros leaving the limiting curve for a three-term recurrence with a more general two-parameter family of initial conditions [17, Thm. 1]. Luong and Tran obtained an explicit numerator-degree bound for the four-term pencil  $1 + s + t + zst$  [16], while Al Hamdani and Tran bounded the number of zeros outside  $(-1, 1)$  for a Chebyshev-type generating function by a conjugate-pair extraction and phase count [1]. These results concern different denominator geometries, so they neither subsume Theorem 1.1 nor are subsumed by it.

Brändén and Leite have proved all-degree real-rootedness and interlacing for the denominator-only Forgács–Tran family through total nonnegativity [5, Thm. 4.5], settling Problem 13 of Forgács and Tran [11]. Their theorem is the case  $N = 1$ ; arbitrary fixed initial conditions can force nonreal or nonpositive zeros, and Theorem 1.1 says that this failure is confined to a bounded defect. In the special case  $(\deg Q, r) = (2, 1)$ , the reduced problem is classical quasi-orthogonality and admits the sharper all-index bound  $\deg B + 2$ ; see Remark 3.3. The general denominator recurrence need not have Favard three-term form.

**1.4. Organization.** Section 2 identifies proper numerators with polynomial initial conditions, constructs the canonical Laurent weight, proves the eventual degree formula, and records the elementary excluded case. Section 3 separates the denominator geometry, contour expansion, and phase control of the principal residue amplitude. Section 4 proves weighted principal-pair dominance, converts it into uniform angular discrepancy, and then proves Theorem 1.1. Section 5 develops the global and local zero laws, including equidistribution, angular rigidity, strong-clock spacing, and isolated exceptional attractors. Section 6 gives the spectral synthesis, discusses sharpness, and closes with the remaining questions.

## 2. CANONICAL LAURENT REDUCTION AND EVENTUAL DEGREE

Throughout the global argument, put

$$D(t, z) = Q(t) + zt^r, \quad d = \deg_t D = \max\{\deg Q, r\}.$$

Apart from the elementary linear case in Proposition 2.6 and the local cancellation result Proposition 5.4, we assume (1.1) with  $Q$  nonconstant, together with  $\max\{\deg Q, r\} > 1$ ; such a pencil is called *admissible*. The condition  $r \geq 1$  is essential to the polynomial formulation, since for  $r = 0$  the coefficients are generally rational in  $z$ . Constant  $Q$  is excluded from the global theorem because then a coefficient polynomial may have a zero of growing multiplicity at the origin, and because  $g = -Q/t^r$  then has no critical point, so that neither  $t_a$  nor  $I_{Q,r}$  exists. For an admissible pencil,  $rQ(t) - tQ'(t)$  has a positive zero, hence  $t_a$ , and at  $r = 1$  one has  $\deg Q \geq 2$  and the negative zero  $t_b$ , so that the Forgács–Tran interval  $I_{Q,r}$  exists; at  $\deg Q = r = 1$  that polynomial reduces to the nonzero constant  $Q(0)$ . The endpoints are recalled in Section 3.

**2.1. Initial conditions.** The numerator carries exactly the same information as the polynomial initial conditions of the denominator recurrence.

**Proposition 2.1.** *Write  $D(t, z) = \sum_{j=0}^d d_j(z)t^j$ , so that  $d_0 = Q(0) \neq 0$ . For each  $N = \sum_{m=0}^{d-1} N_m(z)t^m$  with  $N_m \in \mathbb{R}[z]$ , the coefficients  $P_m$  of  $N/D$  satisfy*

$$\sum_{j=0}^m d_j(z)P_{m-j}(z) = N_m(z) \quad (0 \leq m < d).$$

This is a lower-triangular system with constant diagonal  $d_0 = Q(0)$ , so  $N \mapsto (P_0, \dots, P_{d-1})$  is a bijection from proper numerators, those with  $\deg_t N < d$ , onto  $\mathbb{R}[z]^d$ . A fixed proper  $N$  is thus the same datum as a fixed vector  $(P_0, \dots, P_{d-1})$  of polynomial initial conditions.

*Proof.* Equating coefficients of  $t^m$  in  $N = D \sum_{k \geq 0} P_k t^k$  gives the displayed equations for  $0 \leq m < d$  and  $\sum_{j=0}^d d_j(z) P_{m-j}(z) = 0$  for  $m \geq d$ . The first  $d$  equations express  $(N_0, \dots, N_{d-1})$  from  $(P_0, \dots, P_{d-1})$  by a lower-triangular matrix with constant diagonal  $d_0 = Q(0) \neq 0$ , invertible over  $\mathbb{R}[z]$ ; the remaining equations then determine  $P_m$  for  $m \geq d$ .  $\square$

**2.2. Canonical Laurent reduction.** The denominator curve itself has a canonical Laurent coordinate. Put

$$g(t) = -\frac{Q(t)}{t^r}.$$

Since  $Q(t) = Q(0) + tQ_1(t)$ , the relation  $D = 0$  gives

$$1 = t \left( -\frac{Q_1(t) + z t^{r-1}}{Q(0)} \right),$$

so  $t$  is a unit in the coordinate ring of  $D = 0$ . Consequently

$$(2.1) \quad \mathbb{R}[t, z]/(D) \cong \mathbb{R}[t, t^{-1}], \quad z \mapsto g(t).$$

Indeed, in the quotient  $t$  is invertible and  $z = g(t)$ , while the substitution homomorphism to  $\mathbb{R}[t, t^{-1}]$  has kernel  $(D)$  because  $D$  is primitive and linear in  $z$ . The next lemma is the coefficient-theoretic form of this identification.

**Lemma 2.2** (Canonical Laurent reduction). *Let  $N \in \mathbb{R}[t, z] \setminus \{0\}$  satisfy  $\deg_t N < d$ , and define its restriction to the denominator curve by*

$$(2.2) \quad L_N(t) = N(t, g(t)) \in \mathbb{R}[t, t^{-1}].$$

*Then  $L_N \neq 0$ . There are unique*

$$\lambda_N \in \mathbb{Z}, \quad B_N \in \mathbb{R}[t], \quad B_N(0) \neq 0,$$

*such that*

$$(2.3) \quad L_N(t) = t^{\lambda_N} B_N(t).$$

*Moreover, there is  $S_N \in \mathbb{R}[t, t^{-1}, z]$  with*

$$(2.4) \quad N(t, z) = D(t, z) S_N(t, z) + t^{\lambda_N} B_N(t),$$

*and hence, for all sufficiently large  $m$ ,*

$$(2.5) \quad P_m(z) = [t^{m-\lambda_N}] \frac{B_N(t)}{D(t, z)}.$$

*More quantitatively, put*

$$p_N = \deg_t N, \quad E_N = \deg_z N, \quad q = \deg Q.$$

*Then (2.5) already holds for every*

$$(2.6) \quad m \geq E_N \max\{q - r, 0\},$$

*and*

$$(2.7) \quad \deg B_N \leq p_N + E_N \max\{q, r\}.$$

*Proof.* If  $L_N \equiv 0$ , then  $N(t, z)$ , viewed as a polynomial in  $z$  over  $\mathbb{R}(t)$ , vanishes at  $z = g(t)$ . Since  $D = t^r(z - g(t))$  is linear in  $z$ ,  $D$  divides  $N$  in  $\mathbb{R}(t)[z]$ . The polynomials  $Q(t)$  and  $t^r$  are coprime because  $Q(0) \neq 0$ , so  $D$  is primitive in  $\mathbb{R}[t][z]$ ; Gauss's lemma gives  $D \mid N$  in  $\mathbb{R}[t, z]$ , contradicting  $\deg_t N < d$ . Thus  $L_N \neq 0$ , and its least Laurent exponent gives the unique factorization (2.3).

In  $\mathbb{R}[t, t^{-1}][z]$ , the polynomial  $N(t, z) - L_N(t)$  vanishes at  $z = g(t)$ , hence is divisible by  $z - g(t) = t^{-r} D(t, z)$ . This gives (2.4). Dividing by  $D$ , the Laurent polynomial  $S_N$  contributes no coefficient of  $t^m$  for all sufficiently large  $m$ , which gives (2.5).

For the quantitative bounds, write

$$N(t, z) = \sum_{\beta=0}^{E_N} n_\beta(t) z^\beta.$$

The difference quotient gives explicitly

$$S_N(t, z) = t^{-r} \sum_{\beta=1}^{E_N} n_\beta(t) \sum_{j=0}^{\beta-1} z^{\beta-1-j} g(t)^j.$$

If  $q \leq r$ , then every Laurent exponent of  $S_N$  is at most  $p_N - r \leq -1$ , so (2.5) holds for every  $m \geq 0$ . If  $q > r$ , then  $p_N \leq q - 1$ , and the largest Laurent exponent in the displayed sum is at most

$$p_N - r + (E_N - 1)(q - r) \leq E_N(q - r) - 1.$$

This proves (2.6). Finally,  $t^{rE_N} L_N(t)$  is a polynomial whose degree is at most

$$p_N + rE_N + E_N \max\{q - r, 0\} = p_N + E_N \max\{q, r\}.$$

Since  $t^{rE_N + \lambda_N} B_N = t^{rE_N} L_N$  and  $rE_N + \lambda_N \geq 0$ , (2.7) follows.  $\square$

**Remark 2.3.** The threshold and complexity bounds in Lemma 2.2 are sharp at the level of the displayed degree data. If  $q > r$ , take  $N = t^{q-1} z^{E_N}$  with  $E_N \geq 1$ . In the explicit difference quotient for  $S_N$ , the term with  $j = E_N - 1$  has a nonzero Laurent term of exponent  $E_N(q - r) - 1$ , so the reduction need not hold one index before (2.6). For the reduced-degree bound, fix  $E_N \geq 1$ . If  $q > r$  and  $0 \leq p_N < q$ , take  $N = (1 + t^{p_N}) z^{E_N}$ ; then  $\lambda_N = -rE_N$  and  $B_N = (-1)^{E_N} (1 + t^{p_N}) Q^{E_N}$ , of degree  $p_N + E_N q$ . If  $q \leq r$  and  $0 \leq p_N < r$ , take  $N = t^{p_N} + cz^{E_N}$ , with  $c \neq 0$  chosen to avoid a possible cancellation of the leading terms when  $q = r$  and  $p_N = 0$ ; then  $\lambda_N = -rE_N$  and  $\deg B_N = p_N + rE_N$ . Thus (2.7) is attained in both degree regimes.

Writing  $B_N(t) = \sum_j b_j t^j$  and expanding against (1.2), the reduced tail is therefore the fixed finite linear combination

$$(2.8) \quad P_m(z) = \sum_j b_j H_{m-\lambda_N-j}(z),$$

where  $H_k = 0$  for  $k < 0$ . Thus the proof reduces to one fixed univariate weight. The canonical factorization also separates a pure index shift from the spectral weight: if two proper numerators for the same denominator satisfy

$$L_{N_2}(t) = c t^s L_{N_1}(t), \quad c \neq 0,$$

then their coefficient sequences satisfy

$$P_m^{(2)}(z) = c P_{m-s}^{(1)}(z)$$

for all sufficiently large  $m$ . Thus  $\lambda_N$  controls only the asymptotic index and degree shift, while the zero geometry of the tail is carried, up to a nonzero scalar, by  $B_N$ .

**2.3. Eventual degree.** From now on write

$$(2.9) \quad F_M(z) = [t^M] \frac{B(t)}{Q(t) + zt^r}, \quad B \in \mathbb{R}[t], \quad B(0) \neq 0.$$

**Lemma 2.4.** *For all sufficiently large  $M$ ,*

$$(2.10) \quad \deg F_M = \left\lfloor \frac{M}{r} \right\rfloor.$$

*Proof.* The geometric expansion

$$(2.11) \quad \frac{B(t)}{Q(t) + zt^r} = \sum_{j \geq 0} (-z)^j t^{rj} \frac{B(t)}{Q(t)^{j+1}}$$

gives

$$(2.12) \quad F_M(z) = \sum_{0 \leq j \leq M/r} (-z)^j [t^{M-rj}] \frac{B(t)}{Q(t)^{j+1}}.$$

Hence  $\deg F_M \leq \lfloor M/r \rfloor$ .

Put

$$\ell = \left\lfloor \frac{M}{r} \right\rfloor, \quad s = M - r\ell, \quad 0 \leq s < r.$$

The coefficient of  $z^\ell$  is

$$(2.13) \quad (-1)^\ell [t^s] \frac{B(t)}{Q(t)^{\ell+1}}.$$

Let

$$\Lambda_Q = -\frac{Q'(0)}{Q(0)} = \sum_{j=1}^n \frac{1}{x_j} > 0.$$

For each fixed  $s$ , expanding  $Q(t)^{-(\ell+1)} = Q(0)^{-(\ell+1)} \prod_{j=1}^n (1 - t/x_j)^{-(\ell+1)}$  and extracting  $t^s$  shows that

$$Q(0)^{\ell+1} [t^s] \frac{B(t)}{Q(t)^{\ell+1}}$$

is a polynomial in  $\ell$  of degree at most  $s$ , with leading coefficient  $B(0)\Lambda_Q^s/s!$ . It is therefore nonzero for all sufficiently large  $\ell$ , so the upper degree bound is attained.  $\square$

**Remark 2.5.** For  $B = 1$ , the bound  $\deg F_M \leq \lfloor M/r \rfloor$  is [12, Lem. 8] and is attained at every  $M$ , since  $\prod_j (1 - t/x_j)^{-(\ell+1)}$  has positive coefficients. For a general fixed weight the onset cannot be uniform even when  $\deg B = 1$ . Indeed, take  $Q = 1 - t$ ,  $r = 2$ , and, for any integer  $L \geq 1$ ,

$$B^{(L)}(t) = 1 - (L+1)t.$$

At  $M = 2L + 1$  one has  $\ell = L$ ,  $s = 1$ , and the nominal leading coefficient in (2.13) vanishes because

$$[t] \frac{B^{(L)}(t)}{(1-t)^{L+1}} = (L+1) - (L+1) = 0.$$

Thus  $\deg F_{2L+1} < L$ , giving arbitrarily late degree drops among linear weights, although Lemma 2.4 holds eventually for every fixed  $B^{(L)}$ .

For the original numerator, Lemma 2.2 and Lemma 2.4 give the exact eventual formula

$$(2.14) \quad \deg P_m = \left\lfloor \frac{m - \lambda_N}{r} \right\rfloor,$$

and hence  $\deg P_m = m/r + O_{Q,r,N}(1)$ .

**2.4. The elementary linear case.** The excluded case  $\deg Q = r = 1$  has no Forgács–Tran interval, but the bounded-defect conclusion is immediate.

**Proposition 2.6.** *Suppose  $\deg Q = r = 1$ . Then the first conclusion of Theorem 1.1 holds: there is a  $C$  such that every nonzero  $P_m$  has at most  $C$  zeros outside  $(0, \infty)$ , counted with multiplicity. Its second conclusion is vacuous here,  $I_{Q,r}$  being undefined for  $\deg Q = r = 1$ .*

*Proof.* Write

$$Q(t) = q_0 + q_1 t, \quad q_0 > 0, \quad q_1 < 0.$$

The properness condition  $\deg_t N < 1$  forces  $N(t, z) = R(z)$ . Hence

$$\begin{aligned} \frac{R(z)}{q_0 + (q_1 + z)t} &= \sum_{m \geq 0} \frac{(-1)^m R(z) (q_1 + z)^m}{q_0^{m+1}} t^m, \\ P_m(z) &= \frac{(-1)^m}{q_0^{m+1}} R(z) (z + q_1)^m. \end{aligned}$$

The moving zero  $-q_1$  is positive, and every exceptional zero belongs to the fixed polynomial  $R$ .  $\square$

### 3. SPECTRAL GEOMETRY, RESIDUES, AND THE PRINCIPAL AMPLITUDE

The parameterization, minimum-modulus property, and endpoint expansions come from [12]; the uniform separation estimates and weighted consequences are the ingredients needed below.

#### 3.1. Forgács–Tran geometry and endpoint separation. Let

$$g(t) = -\frac{Q(t)}{t^r}.$$

Following Forgács and Tran, let  $t_a > 0$  be the smallest positive critical point of  $g$ , and set  $b = +\infty$  when  $r > 1$ , while for  $r = 1$  let  $t_b < 0$  be the unique negative critical point of  $g$  [12, Lem. 5] and put  $b = g(t_b)$ . This is the endpoint convention of [12, Thm. 1], where both endpoints are read off the polynomial  $t^{r-1}(rQ(t) - tQ'(t))$ ; its zero of order  $r - 1$  at the origin is what makes the upper endpoint unbounded for  $r > 1$ . Set

$$(3.1) \quad a = g(t_a), \quad I_{Q,r} = (a, b).$$

**Theorem 3.1** (Forgács–Tran geometry and endpoint separation). *There are real-analytic functions*

$$\tau : (0, \pi/r) \rightarrow (0, \infty), \quad z : (0, \pi/r) \rightarrow I_{Q,r},$$

*with  $z$  strictly increasing,  $z(\theta) \rightarrow a$  as  $\theta \downarrow 0$  and  $z(\theta) \rightarrow b$  as  $\theta \uparrow \pi/r$ , such that*

$$(3.2) \quad t_{\pm}(\theta) = \tau(\theta)e^{\pm i\theta}$$

*are zeros of  $Q(t) + z(\theta)t^r$ . They are the only denominator zeros in the closed disk  $|t| \leq \tau(\theta)$ .*

*On every compact subinterval of  $(0, \pi/r)$ , the remaining denominator zeros are separated from the principal pair by a uniform modulus ratio. Near each endpoint, write  $\zeta_j(\theta) = t_j(\theta)/\tau(\theta)$  for the normalized denominator zeros. At  $\theta \downarrow 0$ , a smallest zero  $x_1$  of  $Q$  of multiplicity  $\rho \geq 2$  produces a cluster of  $\rho$  zeros tending to  $x_1$ , whose  $\rho - 2$  nonprincipal members satisfy a linear modulus gap*

$$(3.3) \quad |\zeta_j(\theta)| \geq 1 + c\delta,$$

*for some  $c > 0$ ,  $\delta$  denoting the angular distance to the endpoint, while every zero outside the cluster satisfies*

$$(3.4) \quad |\zeta_j(\theta)| \geq 1 + c_*$$

*for a fixed  $c_* > 0$ . When  $\rho = 1$  the cluster is the principal pair alone and every nonprincipal zero satisfies (3.4). At  $\theta \uparrow \pi/r$  with  $r > 1$ , the finite normalized cluster approaches the  $r$ th roots of  $-1$  and its nonprincipal members satisfy (3.3); if  $\deg Q > r$ , the remaining  $\deg Q - r$  roots tend to infinity. At  $\theta \uparrow \pi$  with  $r = 1$ , the principal pair coalesces at  $t_b$  to a zero of multiplicity exactly two, and every other normalized zero satisfies (3.4).*

*Proof.* The existence, strict monotonicity, and endpoint limits of  $\tau$  and  $z$  are established in Lemmas 2–6 of [12]. The minimum-modulus assertion is [12, Props. 1–2]. Both exclude  $(\deg Q, r) = (2, 1)$ , where  $\tau$  is constant; that case is Remark 3.3. Compact-uniform separation follows by continuity of the roots, compactness, and the strict pointwise modulus gap.

The endpoint expansions are the second and third cases in the proof of [12, Prop. 3]. We extract the uniform separation asserted above. At the lower endpoint, a smallest zero of multiplicity  $\rho \geq 2$  gives normalized cluster roots

$$\zeta_j(\theta) = 1 + \frac{\cos(\pi/\rho) - \omega_j}{\sin(\pi/\rho)} \theta + O(\theta^2), \quad \omega_j = e^{(2j-1)\pi i/\rho},$$

so that

$$|\zeta_j(\theta)| = 1 + \frac{\cos(\pi/\rho) - \Re \omega_j}{\sin(\pi/\rho)} \theta + O(\theta^2);$$

the linear coefficient vanishes exactly for the two principal indices  $\omega_j = e^{\pm i\pi/\rho}$  and is strictly positive otherwise. At the upper endpoint for  $r > 1$ , with  $\eta = \pi/r - \theta$  and  $\omega_j$  now the  $r$ th roots of  $-1$ ,

$$|\zeta_j(\theta)| = 1 + \frac{\cos(\pi/r) - \Re \omega_j}{\sin(\pi/r)} \eta + O(\eta^2),$$

again with vanishing linear coefficient only for the principal pair  $\omega_j = e^{\pm i\pi/r}$ . Since the clusters are finite, the positive linear coefficients have a uniform positive minimum and the remainders are uniform over the cluster, giving (3.3). For  $\rho = 2$  the cluster is the principal pair alone and (3.3) is vacuous.

The zeros outside the cluster keep a fixed gap. At the lower endpoint  $\tau(\theta) \rightarrow t_a$  and  $z(\theta) \rightarrow a$ . Any denominator root that does not remain bounded has modulus tending to infinity and hence, since  $t_a > 0$ , satisfies (3.4) after the endpoint neighborhood is decreased. Among the roots that remain bounded, the normalized values converge to the zeros of the limit polynomial obtained at  $(\tau, z) = (t_a, a)$ , namely  $\zeta = 1$  with multiplicity  $\max\{\rho, 2\}$  together with the ratios  $t_j/t_a$  formed from the remaining bounded zeros  $t_j$  of  $Q(t) + at^r$ , each of modulus exceeding  $t_a$ . The minimum-modulus assertion is quantified over  $\theta \in (0, \pi/r)$  and in the limit gives only  $|t_j| \geq t_a$ , so strictness is supplied directly. Let  $t_e \neq 0$  and  $z_e$  be real with  $Q(t_e) + z_e t_e^r = 0$ . For  $|t| = |t_e|$  with  $t \neq t_e$  one has  $\Re t < t_e$  if  $t_e > 0$  and  $\Re t > t_e$  if  $t_e < 0$ , so

$$|t - x_j|^2 - |t_e - x_j|^2 = 2x_j(t_e - \Re t) \quad (1 \leq j \leq n)$$

has the sign of  $t_e$  for every  $j$ ; hence  $|Q(t)| \neq |Q(t_e)| = |z_e| |t_e|^r = |z_e t_e^r|$  and  $t$  is not a zero of  $Q(t) + z_e t^r$  [12, Prop. 3]. At the lower endpoint this applies with  $t_e = t_a$  and  $z_e = a$ . For  $\rho \geq 2$ , both  $Q$  and  $Q'$  vanish at  $x_1$ , so  $x_1$  is a critical point of  $g$ ; as  $tQ'(t) - rQ(t) < 0$  on  $(0, x_1)$ , it is the smallest positive one. Hence  $t_a = x_1$  and  $a = g(x_1) = -Q(x_1)/x_1^r = 0$ , so the limit polynomial is

$$Q(0)(1 - \zeta)^\rho \prod_{j > \rho} \left(1 - \frac{x_1 \zeta}{x_j}\right),$$

whose zeros off  $\zeta = 1$  are the fixed ratios  $x_j/x_1 > 1$ . Continuity of the zeros in the coefficients gives (3.4) for any  $c_*$  below the least of these ratios diminished by one.

At the upper endpoint for  $r = 1$  the comparison above applies with  $t_e = t_b < 0$  and  $z_e = b$ , the sign in the display reversing with  $t_e$ . A multiplicity above two at  $t_b$  would force  $Q''(t_b) = 0$ , whereas the zeros of  $Q''$  are positive; the collision is therefore exactly double, and  $\deg Q \geq 2 > r$  leaves no escaping zero, the leading  $t$ -coefficient of  $Q(t) + zt$  being the nonzero leading coefficient of  $Q$ , so the remaining  $\deg Q - 2$  zeros satisfy (3.4) by continuity of the zeros in the coefficients [12, Prop. 3]. The upper endpoint for  $r > 1$  is the third case of the same proof. If  $\deg Q > r$ , Rouché's theorem on any fixed circle  $|t| = R$  gives exactly  $r$  zeros inside for all sufficiently large  $z$ , since  $zt^r$  dominates  $Q(t)$  there. Thus every remaining root tends to infinity, and in particular stays outside the fixed separating contours used below.  $\square$

**Lemma 3.2.** *The principal branch*

$$\gamma(\theta) := t_+(\theta) = \tau(\theta)e^{i\theta}$$

*extends to a regular real-analytic arc on the closed parameter interval  $[0, \pi/r]$ , with the upper endpoint interpreted by its finite limit  $\gamma(\pi/r) = 0$  when  $r > 1$ .*

*More precisely, at a finite endpoint let  $\delta$  denote angular distance to the endpoint, let  $t_e \neq 0$  be the limiting principal root, and let  $k \geq 2$  be its multiplicity as a zero of the limiting denominator. Then*

$$(3.5) \quad \gamma(\theta) = t_e + \gamma_e \delta + O(\delta^2), \quad \gamma_e \neq 0,$$

*and*

$$(3.6) \quad z(\theta) - z_e = c_e \delta^k (1 + O(\delta)), \quad c_e \neq 0.$$

*At the upper endpoint when  $r > 1$ , with  $\eta = \pi/r - \theta$ ,*

$$(3.7) \quad \gamma(\theta) = \eta T(\eta), \quad T(0) \neq 0,$$



for a real-analytic complex-valued function  $T$ .

*Proof.* At a finite endpoint write

$$Q(t) + z_e t^r = (t - t_e)^k G(t), \quad G(t_e) \neq 0.$$

A nearby denominator root satisfies

$$(t - t_e)^k = -(z - z_e) \frac{t^r}{G(t)}.$$

On the interior side write  $z - z_e = \pm y^k$ ,  $y \geq 0$ , with the sign fixed. After choosing an analytic  $k$ th root of the nonvanishing factor on the right, the  $k$  local branches have the form

$$t(y) = t_e + \gamma y + O(y^2), \quad \gamma \neq 0.$$

The principal branch is nonreal for interior parameters. Its  $\gamma$  therefore has nonzero imaginary part: if  $\gamma$  were real, conjugation and uniqueness of the local analytic branch would make  $t(y)$  real for real  $y$ . Since  $t_e$  is real and nonzero,

$$\delta = \pm \Im \log \left( \frac{t(y)}{t_e} \right)$$

has nonzero derivative at  $y = 0$ . Inverting  $y$  as a real-analytic function of  $\delta$  gives (3.5), and  $z - z_e = \pm y^k$  gives (3.6).

When  $r > 1$ , the upper endpoint has  $z \rightarrow +\infty$ ; put  $w = z^{-1/r} > 0$ , the positive real root, and take the principal branch of  $Q(t)^{1/r}$  near  $t = 0$ , which is available because  $Q(0) > 0$ . From  $t^r = -Q(t)/z$ , each small root satisfies

$$t = \omega w Q(t)^{1/r}, \quad \omega^r = -1.$$

The implicit-function theorem gives a branch analytic in  $w$  with nonzero derivative at 0. On the principal branch,  $\eta = -r^{-1} \arg Q(t_+)$ , and

$$\frac{d\eta}{dw}(0) = \frac{-Q'(0)}{rQ(0)} Q(0)^{1/r} \sin(\pi/r) > 0,$$

the sign holding because  $Q'(0) = -Q(0) \sum_{j=1}^n x_j^{-1} < 0$ . Thus  $w$  is real-analytic in  $\eta$ , which gives (3.7). The displayed expansions, together with  $\gamma'(\theta) = e^{i\theta}(\tau'(\theta) + i\tau(\theta)) \neq 0$  in the open interval, prove regularity on the closed arc.  $\square$

**Remark 3.3.** For  $\deg Q = 2$  and  $r = 1$ , write  $Q(t) = q_0 + q_1 t + q_2 t^2$ , where  $q_0, q_2 > 0$  and  $q_1 < 0$ . The two denominator roots have product  $q_0/q_2$ , so

$$\tau = \sqrt{q_0/q_2}, \quad z(\theta) = -q_1 - 2\sqrt{q_0 q_2} \cos \theta, \quad 0 < \theta < \pi.$$

Thus  $z$  is strictly increasing,

$$I_{Q,1} = (-q_1 - 2\sqrt{q_0 q_2}, -q_1 + 2\sqrt{q_0 q_2}),$$

the principal pair exhausts the denominator zeros, and both endpoint collisions are exactly double.

This is also the classical quasi-orthogonal case. With

$$p_m(z) = q_0(-q_0)^m H_m(z),$$

coefficient comparison gives

$$p_m = (q_0 q_2)^{m/2} U_m \left( \frac{z + q_1}{2\sqrt{q_0 q_2}} \right).$$

If  $B(t) = \sum_{j=0}^K b_j t^j$  with  $b_0 b_K \neq 0$ , then

$$F_M(z) = \frac{1}{q_0(-q_0)^M} \sum_{j=0}^{\min\{K,M\}} b_j (-q_0)^j p_{M-j}(z).$$

For  $M \geq K$ , Shohat's theorem in the form [7, Lem. 3.1] gives at least  $M - K$  distinct real zeros of odd multiplicity in  $\overline{I_{Q,1}}$ , hence at least  $M - K - 2$  in  $I_{Q,1}$ . Since  $\deg F_M = M$ , at most  $K + 2$  zeros lie outside the open interval, counted with multiplicity; for  $M < K$  the same bound is trivial. Thus the reduced defect is at most  $\deg B + 2$  at every index. The denominator recurrence of a general admissible pencil is not of Favard three-term form.

**3.2. Contour-separated residues.** The core contour-pushing argument is the standard meromorphic coefficient extraction of [10, Thm. IV.10 and Note IV.27]; the following parameter-uniform form records what is needed here for an arbitrary rectifiable Jordan contour, including holomorphy in  $z$  and the bookkeeping at multiple poles.

**Lemma 3.4** (Contour-separated pole expansion). *Let  $r \geq 1$ , let  $Q, B \in \mathbb{C}[t]$  with  $Q(0) \neq 0$ , and set*

$$D(t, z) = Q(t) + zt^r, \quad F_M(z) = [t^M] \frac{B(t)}{D(t, z)}.$$

*Let  $U \subset \mathbb{C}$ , and let  $\Gamma$  be a positively oriented rectifiable Jordan curve whose interior contains 0, such that  $D(t, z) \neq 0$  for  $t \in \Gamma$  and  $z \in U$ . If  $\mathcal{C}(z)$  denotes the set of distinct zeros of  $D(\cdot, z)$  inside  $\Gamma$ , then for every  $M \geq 0$ ,*

$$(3.8) \quad F_M(z) = - \sum_{\xi \in \mathcal{C}(z)} \operatorname{Res}_{t=\xi} \frac{B(t)}{t^{M+1}D(t, z)} + E_M(z), \quad E_M(z) = \frac{1}{2\pi i} \int_{\Gamma} \frac{B(t)}{t^{M+1}D(t, z)} dt.$$

*Each distinct pole contributes its residue once, with the residue at a multiple zero understood in the usual higher-order sense. If every root in  $\mathcal{C}(z)$  is simple, its contribution is*

$$(3.9) \quad - \frac{B(\xi)}{\partial_t D(\xi, z)} \xi^{-M-1}.$$

*If, moreover,*

$$C_{\Gamma} = \sup_{z \in U, t \in \Gamma} \left| \frac{B(t)}{D(t, z)} \right| < \infty, \quad R_{\Gamma} = \inf_{t \in \Gamma} |t| > 0,$$

*then*

$$(3.10) \quad |E_M(z)| \leq \frac{L_{\Gamma} C_{\Gamma}}{2\pi} R_{\Gamma}^{-M-1} \quad (z \in U),$$

*where  $L_{\Gamma}$  is the length of  $\Gamma$ . Normalizing by any positive scale bounded by  $\sigma R_{\Gamma}$  with  $\sigma < 1$  therefore turns (3.10) into a bound of order  $\sigma^M$ . If  $U$  is open, then  $E_M$  is holomorphic in  $z$ .*

*Proof.* Take a small positively oriented circle about the origin containing no zero of  $D(\cdot, z)$ . Cauchy's coefficient formula gives  $F_M$  as the integral of  $B(t)/(t^{M+1}D(t, z))$  on that circle. Expanding the contour to  $\Gamma$  crosses exactly the poles in  $\mathcal{C}(z)$ , and the residue theorem gives (3.8). Formula (3.9) is the residue at a simple zero. The remainder estimate follows by bounding the integrand: on  $\Gamma$  one has  $|B(t)/D(t, z)| \leq C_{\Gamma}$  and  $|t|^{-M-1} \leq R_{\Gamma}^{-M-1}$ , while  $\Gamma$  has length  $L_{\Gamma}$ . Holomorphicity follows by differentiation under the integral sign.  $\square$

**3.3. The principal amplitude and its phase.** For the fixed reduced numerator  $B$ , at any simple denominator zero put

$$(3.11) \quad \mathcal{W}(t, z) = - \frac{B(t)}{\partial_t D(t, z)}.$$

For  $0 < \theta < \pi/r$ , define the weighted principal amplitude by

$$(3.12) \quad W(\theta) = \mathcal{W}(t_+(\theta), z(\theta)).$$

The denominator in (3.11) at  $t = t_+$  equals  $-t_+(\theta)^r g'(t_+(\theta))$ , since  $z(\theta) = -Q(t_+)/t_+^r$  and  $tQ'(t) - rQ(t) = -t^{r+1}g'(t)$ . A double principal root would make  $t_+(\theta)$  a critical point of  $g$ , that is, would force  $t_+Q'(t_+) = rQ(t_+)$ . This cannot occur. The zeros of  $Q$  are positive, so  $Q$  does not vanish off the real axis, and with  $x_1, \dots, x_n$  as in (1.1), listed with multiplicity,

$$\Im \frac{tQ'(t)}{Q(t)} = \Im \sum_{j=1}^n \frac{t}{t - x_j} = -\Im t \sum_{j=1}^n \frac{x_j}{|t - x_j|^2} < 0 \quad (\Im t > 0),$$

whereas  $r$  is real. Since  $t_+(\theta) = \tau(\theta)e^{i\theta}$  with  $0 < \theta < \pi/r \leq \pi$  lies in the open upper half-plane,

$$(3.13) \quad g'(\gamma(\theta)) \neq 0 \quad (0 < \theta < \pi/r),$$

where  $\gamma = t_+$  is the principal branch of Lemma 3.2. The principal root is therefore simple on the open interval, and

$$(3.14) \quad W(\theta) = \frac{B(\gamma(\theta))}{\gamma(\theta)^r g'(\gamma(\theta))}$$

is finite and real-analytic on  $(0, \pi/r)$ .

Remove the two principal residues from the normalized coefficient and define  $R_M$  by the exact identity

$$(3.15) \quad \tau(\theta)^{M+1} F_M(z(\theta)) = 2\Re\left(W(\theta)e^{-i(M+1)\theta}\right) + R_M(\theta).$$

When a contour contains only the principal pair, Lemma 3.4 identifies  $R_M$  with its normalized contour remainder. When a larger endpoint cluster must be retained, the same lemma expresses  $R_M$  as the explicit residues of the other cluster members plus that single contour remainder. Figure 1(A) shows this decomposition for one weight, with an interior zero of the amplitude and the endpoint windows of (4.4).

**Lemma 3.5.** *Let  $K = \deg B$ . The zeros of  $W$  in  $(0, \pi/r)$  are precisely the parameters for which  $B(\gamma(\theta)) = 0$ . If they are  $\theta_1, \dots, \theta_J$ , with multiplicities  $\nu_1, \dots, \nu_J$ , then*

$$(3.16) \quad \sum_{j=1}^J \nu_j \leq K,$$

and near  $\theta_j$ ,

$$(3.17) \quad W(\theta) = (\theta - \theta_j)^{\nu_j} U_j(\theta), \quad U_j(\theta_j) \neq 0.$$

Near either endpoint,

$$(3.18) \quad W(\theta) = \delta^p V(\delta),$$

where  $\delta$  is angular distance to the endpoint,  $p \in \mathbb{Z}$ , and  $V$  is real-analytic and nonvanishing at 0. At a finite endpoint where the limiting principal root has denominator multiplicity  $k$  and  $B$  vanishes there to order  $\nu$ ,

$$p = \nu - (k - 1).$$

Thus at the lower endpoint  $k = \max\{\rho, 2\}$ , where  $\rho$  is the multiplicity of the smallest zero of  $Q$ , and at the finite upper endpoint  $r = 1$  one has  $k = 2$ . At the upper endpoint for  $r > 1$ , one has  $p = 1$ .

Consequently, on every component of  $(0, \pi/r) \setminus \{\theta_1, \dots, \theta_J\}$ , a continuous branch  $\psi = \arg W$  satisfies

$$(3.19) \quad |\psi'(\theta)| \leq \kappa_{Q,r,B}$$

for some finite constant  $\kappa_{Q,r,B}$ .

*Proof.* The principal root is simple in the open interval by (3.13), and

$$\gamma'(\theta) = e^{i\theta}(\tau'(\theta) + i\tau(\theta)) \neq 0.$$

Hence composition with  $\gamma$  preserves the multiplicity of any zero of  $B$ . Since  $\gamma$  is injective, the zeros of  $B \circ \gamma$  have total multiplicity at most  $K$ , proving (3.16) and (3.17).

By (3.14), the endpoint exponent of  $W$  is the order of  $B(\gamma)$  minus that of  $\gamma^r g'(\gamma)$ . At a finite endpoint,  $z_e - g(t)$  vanishes to order  $k$  at  $t_e$ , so  $t^r g'(t)$  vanishes there to order  $k - 1$ . By Lemma 3.2,  $\gamma - t_e = \gamma_e \delta(1 + O(\delta))$ , and the asserted exponent  $p = \nu - (k - 1)$  follows. At the upper endpoint for  $r > 1$ ,  $\gamma = \eta T(\eta)$ ,  $B(\gamma) = B(0)(1 + O(\eta))$ , and

$$\partial_t D(\gamma, z) = Q'(\gamma) - \frac{rQ(\gamma)}{\gamma} = -\frac{rQ(0)}{\gamma}(1 + O(\gamma)),$$

so  $W = \eta V(\eta)$  with  $V(0) \neq 0$ .

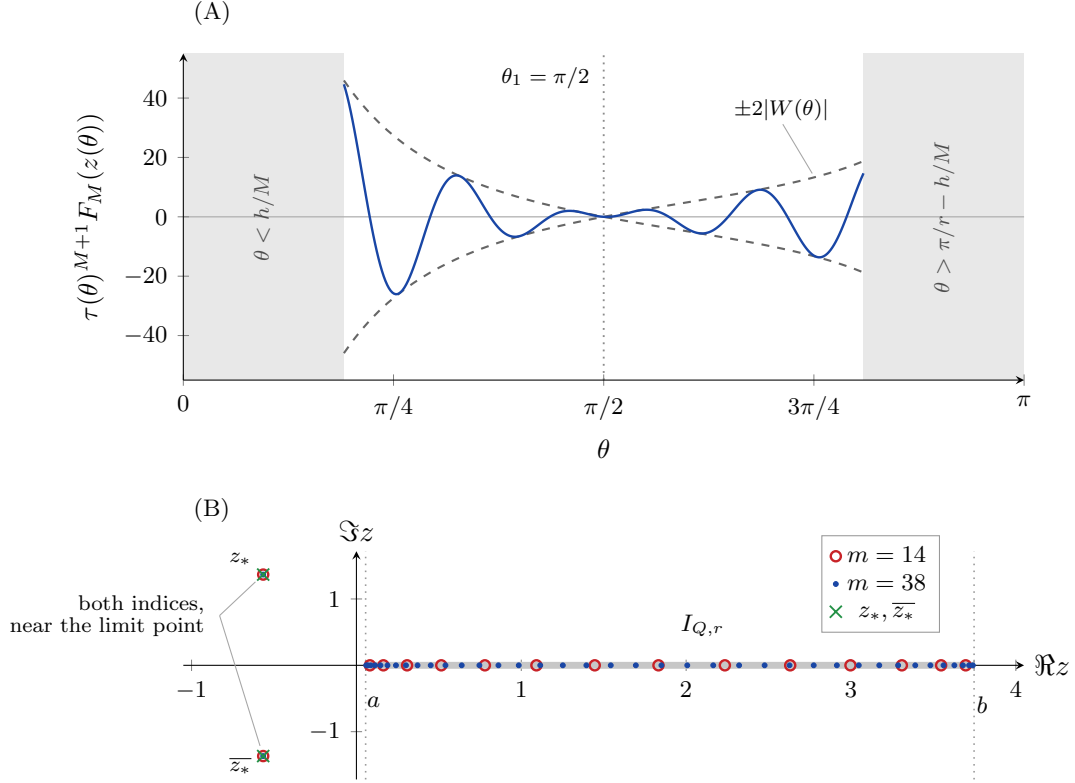


FIGURE 1. The two numerator-sensitive spectral mechanisms, for one denominator  $Q(t) = (1-t)(1-t/2)(1-t/4)$  with  $r = 1$  and  $I_{Q,r} = (0.0559\dots, 3.7466\dots)$ , and one numerator each. **(A)** Co-dominant suppression. For  $B(t) = 7t^2 + 8$  at  $M = 14$ , the normalized coefficient in (3.15) is shown with principal-pair envelope  $\pm 2|W(\theta)|$  dashed. The envelope pinches at  $\theta = \pi/2$ , where  $B(t_+(\pi/2)) = 0$  and the canonical weight suppresses the two-branch oscillation; the exponentially small deleted neighborhood is not visible. The shaded endpoint windows schematically represent (4.4) and are drawn wider than required. **(B)** Uniquely dominant cancellation. Zeros for  $N(t, z) = 1 + z + z^2 + t(2 - z)$  at  $m = 14$  and  $m = 38$ : the positive-real bulk grows while the exceptional conjugate pair is already extremely close to the algebraic limit points  $z_*, \bar{z}_*$ . Corollary 5.6 proves  $\deg P_m = m + 2$  for every  $m$  and  $z_m = z_* + O(3^{-m})$  for that pair.

Finally, near an interior zero,

$$\frac{W'}{W} = \frac{\nu_j}{\theta - \theta_j} + O(1),$$

and near an endpoint,

$$\frac{d}{d\delta} \log W = \frac{p}{\delta} + O(1).$$

The singular terms are real on each side, hence disappear from  $\Im(W'/W) = \psi'$ . Compactness of the remaining subarcs gives (3.19).  $\square$

The singular parts of  $W'/W$  are real: amplitude zeros and endpoint powers change the size of  $W$ , but introduce no singular term into  $\psi' = \Im(W'/W)$ .

The resulting phase variation can be bounded uniformly over all numerator weights of a given degree; the geometric input is Radon's viewing-angle bound.

**Lemma 3.6** (Radon's viewing-angle bound). *Let  $\gamma : [s_0, s_1] \rightarrow \mathbb{C}$ ,  $s \mapsto \gamma(s)$ , be an injective regular real-analytic arc, real-analytic on a neighborhood of the closed interval. Choose a continuous tangent angle  $\vartheta = \arg \gamma'$  and put*

$$\mathcal{K}_\gamma = \text{Var}_{[s_0, s_1]} \vartheta.$$

Then for every  $\beta \in \mathbb{C}$ ,

$$(3.20) \quad \sum_{\mathcal{I}} \text{Var}_{\mathcal{I}} \arg(\gamma - \beta) \leq \mathcal{K}_{\gamma} + \pi,$$

where the sum is over the components  $\mathcal{I}$  of  $[s_0, s_1] \setminus \gamma^{-1}(\{\beta\})$ , and a continuous branch of the argument is used on each component.

*Proof.* This is Radon’s theorem on the viewing angle of an arc of bounded rotation, together with the identification of total rotation with tangent-angle variation; see [9, Lems. 4.39–4.40]. For  $\beta \notin \gamma([s_0, s_1])$ , the viewing angle is a continuous argument of  $\gamma - \beta$ , and Radon’s bound is exactly  $\mathcal{K}_{\gamma} + \pi$ . If  $\beta$  lies on the arc, injectivity gives a unique preimage. Radon’s one-sided viewing angle at that point gives continuous argument branches on the complementary components, and its variation over  $[s_0, s_1]$  bounds the sum in (3.20). Since the total rotation of a regular arc is the variation of its tangent angle, the same bound follows.  $\square$

**Corollary 3.7.** *There are constants  $\kappa_0, \kappa_1 < \infty$ , depending only on the denominator, such that for every  $B \in \mathbb{R}[t]$  with  $B(0) \neq 0$ ,*

$$(3.21) \quad \sum_{\mathcal{I}} \text{Var}_{\mathcal{I}} \arg W \leq \kappa_0 + \kappa_1 \deg B,$$

where the sum is over the components  $\mathcal{I}$  of  $(0, \pi/r) \setminus \{W = 0\}$ . One may take  $\kappa_1 = \mathcal{K}_{\gamma} + \pi$ , where  $\mathcal{K}_{\gamma}$  is the total tangent-angle variation of the principal arc.

*Proof.* By Lemma 3.2, the principal arc  $\gamma(\theta) = t_+(\theta)$  is a regular real-analytic arc on the closed parameter interval. Factor over  $\mathbb{C}$ ,

$$B(t) = b \prod_{\ell=1}^K (t - \beta_{\ell}), \quad K = \deg B.$$

By Lemma 3.5, deleting the zeros of  $W$  is exactly deleting the parameters where  $B \circ \gamma$  vanishes. On each remaining component, choose continuous arguments for the factors of  $B$ . Then

$$\arg B(\gamma(\theta)) = \arg b + \sum_{\ell=1}^K \arg(\gamma(\theta) - \beta_{\ell})$$

up to an additive constant, so its variation is at most the sum of the factor variations. For fixed  $\ell$ , the remaining components refine those of  $[0, \pi/r] \setminus \gamma^{-1}(\{\beta_{\ell}\})$ ; hence their summed variation is bounded by the left side of (3.20). Thus Lemma 3.6 bounds the  $\ell$ th contribution by  $\mathcal{K}_{\gamma} + \pi$ , and summing over  $\ell$  gives  $K(\mathcal{K}_{\gamma} + \pi)$ . The remaining factor in (3.14),

$$(\gamma(\theta)^r g'(\gamma(\theta)))^{-1},$$

depends only on  $Q, r$ . It has finite total phase variation: it is real-analytic and nonvanishing in the open interval, and by Lemma 3.2 together with (3.18) for  $B = 1$  it differs near each endpoint from a real-analytic nonvanishing function by the positive factor  $\delta^p$  alone, which contributes nothing to the argument. A continuous branch of that argument is therefore real-analytic on the closed interval, hence of finite variation. Adding this fixed variation proves (3.21).  $\square$

#### 4. PRINCIPAL-PAIR DOMINANCE AND THE FIXED-NUMERATOR THEOREM

**4.1. Weighted principal-pair dominance.** We now extend the denominator-only estimates of [12, Prop. 3] to the fixed polynomial weight  $B$ . The proof has four pieces. On compact interior subarcs the principal pair has a fixed modulus gap from every other pole. Near an endpoint that gap may close linearly, but outside an angular window of width  $O(M^{-1})$  this still gives fixed exponential suppression. Near a zero of the principal amplitude, an exponentially small deleted interval restores dominance. Every pole outside the retained endpoint cluster is handled at once by the contour remainder.

**Lemma 4.1.** *Let  $c, \varepsilon, C_W > 0$ , let  $M \geq 0$  be an integer, and let  $\delta \in (0, \varepsilon]$ . Suppose  $W(\delta) \neq 0$  and that finitely many pairs  $W_j(\delta), \zeta_j(\delta) \in \mathbb{C}$  satisfy*

$$|W_j(\delta)| \leq C_W |W(\delta)|, \quad |\zeta_j(\delta)| \geq 1 + c\delta.$$

*Then, with  $c' = c/(1 + c\varepsilon)$ ,*

$$(4.1) \quad \frac{1}{|W(\delta)|} \left| \sum_j W_j(\delta) \zeta_j(\delta)^{-M-1} \right| \leq C_W \#\{j\} e^{-c'M\delta}.$$

*In particular, for  $h > 0$  the right side is at most  $C_W \#\{j\} e^{-c'h}$  whenever  $M\delta \geq h$ .*

*Proof.* The inequality  $\log(1+x) \geq x/(1+x)$  for  $x \geq 0$  gives

$$|\zeta_j|^{-M-1} \leq (1 + c\delta)^{-M-1} \leq e^{-c'M\delta}.$$

Summing the residue bounds proves (4.1).  $\square$

**Theorem 4.2** (Weighted principal-pair dominance). *There is a constant  $h = h(Q, r) > 0$  with the following property. For every fixed  $B$ , there are  $M_0 = M_0(Q, r, B)$  and  $c = c(Q, r, B) > 0$ . For  $M \geq M_0$ , define*

$$(4.2) \quad \Theta_{j,M} = \left\{ \theta : |\theta - \theta_j| < e^{-cM/\nu_j} \right\}$$

*for the zeros in Lemma 3.5. Then*

$$(4.3) \quad |R_M(\theta)| \leq \frac{1}{2} |W(\theta)|$$

*whenever*

$$(4.4) \quad \frac{h}{M} \leq \theta \leq \frac{\pi}{r} - \frac{h}{M}, \quad \theta \notin \bigcup_{j=1}^J \Theta_{j,M}.$$

*The total length of the deleted amplitude intervals is exponentially small.*

*Proof.* Fix first a denominator-only endpoint radius  $\varepsilon_* \in (0, \pi/(2r))$  on which the endpoint expansions and the linear modulus-gap estimates of Theorem 3.1 hold. For the fixed weight  $B$ , decrease to some  $0 < \varepsilon \leq \varepsilon_*$  so that every zero  $\theta_j$  of Lemma 3.5 lies in  $(\varepsilon, \pi/r - \varepsilon)$ ; then  $W$  has no zero in  $(0, \varepsilon]$  or in  $[\pi/r - \varepsilon, \pi/r)$ . We use Lemma 3.4 with only the roots whose modulus can compete with the principal pair retained inside the contour. On the compact interior, the uniform minimum-modulus gap and a finite cover give fixed local contours containing only the principal pair. Near the lower endpoint a fixed circle contains the principal pair when the smallest zero is simple and the full  $\rho$ -root endpoint cluster when it has multiplicity  $\rho > 1$ . Near the upper endpoint it contains the principal pair for  $r = 1$  and the  $r$ -root small cluster for  $r > 1$ . The fixed-gap assertions of Theorem 3.1 separate those clusters from every remaining finite root; roots that leave every bounded set remain outside the same fixed circle.

After decreasing  $\varepsilon$ , the hypotheses of (3.10) are therefore uniform on each of these regions. At the unbounded upper endpoint  $r > 1$ , if the separating circle is  $|t| = R$ , then  $z(\theta)R^r$  dominates  $Q(t)$  uniformly on the circle for  $\theta$  sufficiently close to  $\pi/r$ , so the required bound on  $B/D$  is uniform there as well. In particular, after normalization by  $\tau^{M+1}$ , every contour remainder is  $O(\sigma^M)$  for some region-dependent  $\sigma < 1$ . We shall also use repeatedly that at any nonzero denominator root

$$(4.5) \quad \partial_t D(t, z) = Q'(t) - \frac{rQ(t)}{t}.$$

*Compact interior.* On  $\theta \in [\varepsilon, \pi/r - \varepsilon]$ , the retained cluster is the principal pair alone, so Lemma 3.4 and (3.15) give

$$(4.6) \quad |R_M(\theta)| \leq C_I \sigma_I^M, \quad \varepsilon \leq \theta \leq \pi/r - \varepsilon,$$

for some  $C_I < \infty$  and  $\sigma_I < 1$ . Choose

$$c = \frac{1}{2} \log \frac{1}{\sigma_I} > 0.$$

Choose pairwise disjoint fixed neighborhoods of the finitely many  $\theta_j$ . By (3.17), after increasing  $M_0$  so that every  $\Theta_{j,M}$  lies in its fixed neighborhood, there is a  $C_A > 0$  such that throughout the retained compact interior,

$$|W(\theta)| \geq C_A e^{-cM}.$$

The factor  $1/\nu_j$  in the deletion radius is what makes this lower bound independent of the multiplicity of the amplitude zero. Indeed, in the neighborhood of  $\theta_j$  one has  $|W(\theta)| \geq C_j |\theta - \theta_j|^{\nu_j}$ , while away from all those neighborhoods  $|W|$  has a positive minimum. Hence

$$(4.7) \quad \frac{|R_M(\theta)|}{|W(\theta)|} \leq \frac{C_I}{C_A} \sigma_I^{M/2} = o(1)$$

uniformly on the retained compact interior.

For the endpoint regions it is useful to record once the comparison between a fixed-contour error and the principal amplitude. If  $\delta$  is the angular distance to the endpoint, Lemma 3.5 gives  $|W(\theta)| \asymp \delta^p$  for a fixed integer  $p$ . Thus any normalized contour error  $\mathcal{E}_M(\theta)$  satisfying  $|\mathcal{E}_M(\theta)| \leq C\sigma^M$  also satisfies, for every fixed  $h > 0$ ,

$$(4.8) \quad \frac{|\mathcal{E}_M(\theta)|}{|W(\theta)|} \leq C_h \sigma^M M^{\max\{p,0\}} = o(1) \quad (\delta \geq h/M),$$

uniformly in the endpoint region.

*The lower endpoint,  $\theta \downarrow 0$ .* If the smallest zero of  $Q$  is simple, the retained endpoint cluster is the principal pair alone. Hence  $R_M$  is just the contour error and (4.8) gives

$$\frac{|R_M(\theta)|}{|W(\theta)|} = o(1) \quad (h/M \leq \theta \leq \varepsilon).$$

This includes without distinction any parameter at which a nonprincipal root escapes to infinity.

Now let the smallest zero  $x_1$  have multiplicity  $\rho > 1$ . The endpoint analysis of [12, Prop. 3] produces a cluster of actual roots  $t_j(\theta) \rightarrow x_1$  with

$$(4.9) \quad t_j(\theta) = x_1 + \alpha_j \theta + O(\theta^2), \quad \alpha_j = -\frac{x_1 \omega_j}{\sin(\pi/\rho)}, \quad \omega_j = e^{(2j-1)\pi i/\rho}.$$

The principal upper branch is the index with  $\omega_+ = e^{-i\pi/\rho}$ , for which  $t_+ = \tau e^{i\theta}$ ; write  $\alpha_+$  for its coefficient. The  $\alpha_j$  are distinct and nonzero, so all members of this cluster are simple and pairwise distinct for sufficiently small  $\theta > 0$ . Also  $|\alpha_j| = x_1/\sin(\pi/\rho)$  for every  $j$ . Together with  $\tau(\theta) = x_1(1 - \cot(\pi/\rho)\theta) + O(\theta^2)$  this gives  $|\zeta_{\pm}(\theta)| = 1$  for the principal pair and

$$(4.10) \quad |\zeta_j(\theta)| \geq 1 + c_0 \theta$$

for every other cluster member, after decreasing  $\varepsilon$  if necessary.

Let  $\nu = \text{ord}_{x_1} B \geq 0$ , and for a cluster member put

$$W_j(\theta) = \mathcal{W}(t_j(\theta), z(\theta)).$$

Thus  $W_+ = W$ . From (4.9),

$$B(t_j) = \frac{B^{(\nu)}(x_1)}{\nu!} \alpha_j^\nu \theta^\nu (1 + O(\theta)).$$

By (4.5) and Taylor expansion of  $Q$  at  $x_1$ ,

$$Q'(t_j) - \frac{rQ(t_j)}{t_j} = \frac{Q^{(\rho)}(x_1)}{(\rho-1)!} \alpha_j^{\rho-1} \theta^{\rho-1} (1 + O(\theta)).$$

Consequently

$$(4.11) \quad \frac{W_j(\theta)}{W(\theta)} = \left( \frac{\alpha_j}{\alpha_+} \right)^{\nu-\rho+1} (1 + O(\theta)), \quad \left| \frac{W_j(\theta)}{W(\theta)} \right| \rightarrow 1.$$

The error is uniform over the finite cluster, so after decreasing  $\varepsilon$  we may assume  $|W_j| \leq 2|W|$  for every member.

The contour now retains the full  $\rho$ -root cluster. Thus Lemma 3.4 gives

$$R_M(\theta) = \sum_{j \neq +, -} W_j(\theta) \zeta_j(\theta)^{-M-1} + \mathcal{E}_M(\theta), \quad |\mathcal{E}_M(\theta)| \leq C_L \sigma_L^M.$$

Applying Lemma 4.1 with  $\delta = \theta$ ,  $C_W = 2$ , and  $c = c_0$ , put  $c'_0 = c_0/(1 + c_0\varepsilon)$ . On  $\theta \geq h/M$ ,

$$(4.12) \quad \frac{1}{|W(\theta)|} \left| \sum_{j \neq +, -} W_j(\theta) \zeta_j(\theta)^{-M-1} \right| \leq 2(\rho - 2)e^{-c'_0 h}.$$

For  $\rho = 2$  the sum is absent, while (4.8) makes  $|\mathcal{E}_M|/|W| = o(1)$ .

The upper endpoint,  $\theta \uparrow \pi/r$ . When  $r = 1$ , the retained endpoint cluster is again the principal pair alone, so (4.8), with  $\delta = \pi - \theta$ , gives

$$\frac{|R_M(\theta)|}{|W(\theta)|} = o(1) \quad (h/M \leq \pi - \theta \leq \varepsilon).$$

A zero of  $B$  at the collision point is already included in the endpoint exponent  $p$ .

Assume  $r > 1$  and set  $\eta = \pi/r - \theta$ . By Theorem 3.1, the retained small cluster approaches the  $r$ th roots of  $-1$ , the principal roots approaching  $e^{\pm i\pi/r}$ , and every other member satisfies

$$|\zeta_j(\theta)| \geq 1 + c_1\eta$$

for some  $c_1 > 0$ . The limiting normalized roots are distinct, so the small cluster is simple for sufficiently small  $\eta > 0$ . Using (4.5),  $t_j = \tau\zeta_j$ , and  $B(0) \neq 0$ ,

$$\partial_t D(t_j, z(\theta)) = -\frac{rQ(0)}{\tau\zeta_j}(1 + O(\tau)),$$

so with  $W_j = \mathcal{W}(t_j, z(\theta))$ ,

$$(4.13) \quad W_j(\theta) = \frac{B(0)}{rQ(0)} \tau\zeta_j(1 + O(\tau)), \quad \frac{W_j(\theta)}{W(\theta)} = \frac{\zeta_j(\theta)}{\zeta_+(\theta)}(1 + O(\tau)).$$

Hence  $|W_j/W| \rightarrow 1$  uniformly over the finite cluster; after decreasing  $\varepsilon$ , use  $|W_j| \leq 2|W|$ . The contour-separated expansion gives

$$R_M(\theta) = \sum_{j \neq +, -} W_j(\theta) \zeta_j(\theta)^{-M-1} + \mathcal{E}_M(\theta), \quad |\mathcal{E}_M(\theta)| \leq C_U \sigma_U^M.$$

Applying Lemma 4.1 with  $\delta = \eta$ ,  $C_W = 2$ , and  $c = c_1$ , put  $c'_1 = c_1/(1 + c_1\varepsilon)$ . On  $\eta \geq h/M$ ,

$$(4.14) \quad \frac{1}{|W(\theta)|} \left| \sum_{j \neq +, -} W_j(\theta) \zeta_j(\theta)^{-M-1} \right| \leq 2(r - 2)e^{-c'_1 h}.$$

For  $r = 2$  the sum is absent, and (4.8) again gives  $|\mathcal{E}_M|/|W| = o(1)$ .

It remains only to choose the constants, in an order that keeps  $h$  free of  $B$ : first the endpoint rates, then  $h$ , then the  $B$ -dependent threshold  $M_0$ . Because  $\varepsilon \leq \varepsilon_*$ ,

$$c'_0 \geq \frac{c_0}{1 + c_0\varepsilon_*}, \quad c'_1 \geq \frac{c_1}{1 + c_1\varepsilon_*}$$

whenever the corresponding endpoint cluster is present. These lower bounds depend only on  $Q, r$ . Choose  $h = h(Q, r)$  so large that

$$2(\rho - 2)e^{-c'_0 h} \leq \frac{1}{4} \quad (\rho > 1), \quad 2(r - 2)e^{-c'_1 h} \leq \frac{1}{4} \quad (r > 1),$$

which is possible using those denominator-only lower bounds. Then choose  $M_0$  sufficiently large that  $h/M_0 < \varepsilon$ , that (4.7) is at most  $1/2$ , and that every endpoint contour error in (4.8) is at



most  $1/4$ . Combining the explicit near-cluster terms with the contour remainders gives (4.3) throughout (4.4). Finally,

$$\sum_{j=1}^J |\Theta_{j,M}| \leq 2 \sum_{j=1}^J e^{-cM/\nu_j},$$

which is exponentially small because the number and multiplicities of the amplitude zeros are fixed.  $\square$

**Remark 4.3.** The proof does not require the exact sign of the full weighted endpoint cluster. Outside angular windows of width  $h/M$ , the linear endpoint modulus gap becomes the fixed exponential suppression  $e^{-c'_0 h}$  at the lower endpoint and  $e^{-c'_1 h}$  at the upper.

**4.2. Angular discrepancy and proof of the main theorem.** The dominance theorem gives more than a global bounded defect: the zeros have uniformly bounded discrepancy in the angular coordinate. The resulting quantitative statement will imply both Theorem 1.1 and the limiting measure in Proposition 5.1.

For  $0 \leq \alpha < \beta \leq \pi/r$ , put

$$(4.15) \quad I_{\alpha,\beta} = \{z(\theta) : \alpha < \theta < \beta\} \subset I_{Q,r},$$

and let  $Z_M(\alpha, \beta)$  denote the number of zeros of  $F_M$  in  $I_{\alpha,\beta}$ , counted with multiplicity. Here “angular” refers to the Forgács–Tran spectral parameter  $\theta$ , not to the classical Erdős–Turán discrepancy of the complex arguments of polynomial zeros [8, Thm. I].

**Proposition 4.4** (Uniform angular zero discrepancy). *There are constants  $C_0 = C_0(Q, r)$  and  $C_1 = C_1(Q, r)$  such that, for every  $B \in \mathbb{R}[t]$  with  $B(0) \neq 0$ , writing  $K = \deg B$ , all sufficiently large  $M$  satisfy, uniformly for  $0 \leq \alpha < \beta \leq \pi/r$ ,*

$$(4.16) \quad \#\text{dist}\{x \in I_{\alpha,\beta} : F_M(x) = 0\} \geq \frac{M+1}{\pi}(\beta - \alpha) - C_0 - C_1 K,$$

and

$$(4.17) \quad \left| Z_M(\alpha, \beta) - \frac{M+1}{\pi}(\beta - \alpha) \right| \leq C_0 + C_1 K.$$

The threshold in  $M$  may depend on  $B$ , but the discrepancy constants do not.

*Proof.* Let  $h = h(Q, r)$ ,  $c$ , and  $M_0$  be as in Theorem 4.2, and define

$$(4.18) \quad \Omega_M = \left[ \frac{h}{M}, \frac{\pi}{r} - \frac{h}{M} \right] \setminus \bigcup_{j=1}^J \Theta_{j,M}.$$

By (3.16),  $J \leq K$ . Since the amplitude windows are exponentially small, after increasing the  $B$ -dependent threshold we may assume

$$(4.19) \quad (M+1) \sum_{j=1}^J |\Theta_{j,M}| \leq 1.$$

Fix  $\alpha < \beta$ . On each component of  $\Omega_M \cap (\alpha, \beta)$ , choose a continuous branch  $\psi = \arg W$  and put

$$(4.20) \quad \Phi_M(\theta) = (M+1)\theta - \psi(\theta).$$

By (3.19),  $\Phi'_M > 0$  for all sufficiently large  $M$ . At a point where  $\Phi_M = k\pi$ , the principal term in (3.15) has sign  $(-1)^k$ , and Theorem 4.2 gives the same sign for the complete coefficient. Consecutive phase points therefore enclose distinct zeros.

On a component  $\mathcal{I}$  of  $\Omega_M \cap (\alpha, \beta)$ , the increase of  $\Phi_M$  is at least  $(M+1)|\mathcal{I}| - \text{Var}_{\mathcal{I}} \psi$ , so  $\Phi_M$  meets  $\pi\mathbb{Z}$  at least  $((M+1)|\mathcal{I}| - \text{Var}_{\mathcal{I}} \psi)/\pi - 1$  times, and  $\mathcal{I}$  carries at least one zero fewer. The number of components is at most  $J+1$ , so summing the phase count gives

$$\#\text{dist}\{x \in I_{\alpha,\beta} : F_M(x) = 0\} \geq \frac{M+1}{\pi} |\Omega_M \cap (\alpha, \beta)| - \frac{1}{\pi} \sum_{\mathcal{I}} \text{Var}_{\mathcal{I}} \psi - 2(J+1).$$

The length deleted from  $(\alpha, \beta)$  is at most  $2h/M + \sum_j |\Theta_{j,M}|$ . Using (4.19), Corollary 3.7, and  $J \leq K$ , we obtain (4.16) with constants depending only on  $Q, r$ .

For the upper bound, apply (4.16) to the two complementary angular intervals  $(0, \alpha)$  and  $(\beta, \pi/r)$ , omitting either interval when it is empty. Together they contain at least

$$\frac{M+1}{\pi} \left( \frac{\pi}{r} - (\beta - \alpha) \right) - 2(C_0 + C_1 K)$$

distinct zeros outside  $I_{\alpha, \beta}$ , after enlarging the constants once. Since  $\deg F_M = \lfloor M/r \rfloor$  eventually,

$$Z_M(\alpha, \beta) \leq \left\lfloor \frac{M}{r} \right\rfloor - \frac{M+1}{\pi} \left( \frac{\pi}{r} - (\beta - \alpha) \right) + 2(C_0 + C_1 K) \leq \frac{M+1}{\pi} (\beta - \alpha) + 2(C_0 + C_1 K).$$

The distinct-zero lower bound also gives the corresponding lower bound for  $Z_M$ . Renaming the constants proves (4.17).  $\square$

*Proof of Theorem 1.1.* Let

$$L_N(t) = t^{\lambda_N} B_N(t)$$

be the canonical Laurent restriction from Lemma 2.2. For all sufficiently large  $m$ ,

$$P_m(z) = F_{m-\lambda_N}(z)$$

with the fixed weight  $B_N$ . Apply Proposition 4.4 to the full angular interval  $(0, \pi/r)$ . Together with Lemma 2.4, it gives, for all sufficiently large  $m$ ,

$$\#_{\text{dist}} \{x \in I_{Q,r} : P_m(x) = 0\} \geq \deg P_m - O_{Q,r,N}(1),$$

and hence only  $O_{Q,r,N}(1)$  remaining zeros counted with multiplicity. More precisely, the constants in (4.16) depend on the weight only through  $\deg B_N$ , so after increasing the denominator-only constant by one, the eventual defect is at most  $C_0 + C_1 \deg B_N$ . Since  $I_{Q,r} \subset (0, \infty)$ , the same qualitative bound holds outside the positive ray. The exact eventual degree formula is (2.14). Enlarging the all-index constant to cover the finitely many preceding nonzero coefficient polynomials proves the theorem.  $\square$

## 5. GLOBAL AND LOCAL ZERO LAWS

**5.1. Global zero distribution and angular rigidity.** After normalization by degree, the zeros of every fixed numerator converge weakly to a common limiting measure determined by the denominator. For  $P \in \mathbb{R}[z]$  of positive degree  $k$ , with zeros  $\xi_1, \dots, \xi_k$  repeated by multiplicity, write

$$\mu[P] = \frac{1}{k} \sum_{i=1}^k \delta_{\xi_i}.$$

**Proposition 5.1.** *Let  $z : (0, \pi/r) \rightarrow I_{Q,r}$  be the parameterization of Theorem 3.1, let  $\mu_\infty$  be the pushforward of  $\frac{r}{\pi} d\theta$  under  $z$ , and write*

$$N\left(t, -\frac{Q(t)}{t^r}\right) = t^{\lambda_N} B_N(t), \quad B_N(0) \neq 0,$$

*for the canonical Laurent restriction. There are constants  $D_0 = D_0(Q, r)$  and  $D_1 = D_1(Q, r)$  such that, for all sufficiently large  $m$ ,*

$$(5.1) \quad \sup_{0 \leq \alpha < \beta \leq \pi/r} \left| \mu[P_m](I_{\alpha, \beta}) - \frac{r}{\pi} (\beta - \alpha) \right| \leq \frac{D_0 + D_1 \deg B_N}{\deg P_m}.$$

*Consequently  $\mu[P_m] \Rightarrow \mu_\infty$  along the coefficient polynomials of positive degree. In particular, every fixed numerator has the same limiting zero distribution as the denominator-only sequence  $H_m$ .*

*Proof.* Put  $M = m - \lambda_N$ , so that for all sufficiently large  $m$ ,  $P_m = F_M$  with weight  $B_N$ , and put  $d_M = \deg P_m = \lfloor M/r \rfloor$ . By Proposition 4.4, with  $K = \deg B_N$ ,

$$\left| Z_M(\alpha, \beta) - \frac{M+1}{\pi}(\beta - \alpha) \right| \leq C_0 + C_1 K.$$

Write  $M = rd_M + s$  with  $0 \leq s < r$ . Since  $0 \leq \beta - \alpha \leq \pi/r$ ,

$$\frac{\beta - \alpha}{\pi} \left| \frac{M+1}{d_M} - r \right| = \frac{\beta - \alpha}{\pi} \frac{s+1}{d_M} \leq \frac{1}{d_M}.$$

Dividing the discrepancy estimate by  $d_M$  therefore proves (5.1), after increasing the denominator-only constant by one.

For weak convergence, let  $0 < \alpha < \beta < \pi/r$ . By (5.1) the compact interval  $z([\alpha, \beta])$  carries  $\mu[P_m]$ -mass at least  $\frac{\pi}{\pi}(\beta - \alpha) - (D_0 + D_1 \deg B_N)/\deg P_m$ , which exceeds any prescribed bound below 1 once  $\beta - \alpha$  is close enough to  $\pi/r$  and  $m$  is large; the sequence is therefore tight.

The same estimate gives  $\mu[P_m](I_{\alpha, \beta}) \rightarrow \mu_\infty(I_{\alpha, \beta})$  for every such  $\alpha, \beta$ , and  $\mu_\infty$  has no atoms because  $z$  is strictly monotone, so these intervals form a convergence-determining class.  $\square$

**Corollary 5.2.** *For all sufficiently large  $m$ , list the zeros of  $P_m$  in  $I_{Q,r}$ , counted with multiplicity, as*

$$x_{1,m} \leq \cdots \leq x_{s_m,m}, \quad \theta_{j,m} = z^{-1}(x_{j,m}),$$

and put  $M = m - \lambda_N$ . There are constants  $E_0 = E_0(Q, r)$ ,  $E_1 = E_1(Q, r)$  such that

$$(5.2) \quad \left| \theta_{j,m} - \frac{\pi j}{M+1} \right| \leq \frac{\pi(E_0 + E_1 \deg B_N)}{M+1} \quad (1 \leq j \leq s_m).$$

Thus every bulk-zero angle lies within  $O_{Q,r,N}(m^{-1})$  of the uniform angular clock.

*Proof.* Let  $\Delta_B = C_0 + C_1 \deg B_N$  be the discrepancy constant in Proposition 4.4. Fix a zero angle  $\theta$ , and let  $j_-$  and  $j_+$  be respectively the numbers of interval zeros with angle strictly less than, and at most,  $\theta$ . Applying (4.17) to  $(0, \theta)$ , and then to  $(0, \theta + \varepsilon)$  followed by  $\varepsilon \downarrow 0$ , gives

$$\left| j_- - \frac{M+1}{\pi} \theta \right| \leq \Delta_B, \quad \left| j_+ - \frac{M+1}{\pi} \theta \right| \leq \Delta_B.$$

Every index  $j$  belonging to that multiple zero satisfies  $j_- < j \leq j_+$ , hence

$$\left| j - \frac{M+1}{\pi} \theta \right| \leq \Delta_B + 1.$$

Absorb the final 1 into the denominator-only constant and rearrange.  $\square$

**5.2. Local phase quantization and strong-clock spacing.** The global localization in Corollary 5.2 is reminiscent of clock behavior for orthogonal-polynomial zeros [15, p. 488], but by itself it is weaker than strong-clock spacing: the error in (5.2) is of the same order  $M^{-1}$  as the nominal spacing. Exponentially accurate finite-size zero-location laws from two-term spectral competition also appear in a different setting in [3]. On zero-free compact subarcs here, the phase law is much more rigid.

**Proposition 5.3.** *Fix  $B \in \mathbb{R}[t]$  with  $B(0) \neq 0$ , put*

$$F_M(z) = [t^M] \frac{B(t)}{Q(t) + zt^r},$$

and let  $W$  be the principal amplitude (3.12). Let  $\mathcal{J}$  be a compact interval contained in one component of the complement in  $(0, \pi/r)$  of the amplitude zeros of Lemma 3.5, and choose a continuous phase  $\psi = \arg W$  there. Then there are  $\sigma \in (0, 1)$  and  $M_0$  such that, for  $M \geq M_0$ , every zero of  $F_M$  in  $z(\mathcal{J})$  is simple and is  $z(\theta_{k,M})$  for a unique  $\theta_{k,M} \in \mathcal{J}$ , the indices  $k$  running consecutively, with

$$(5.3) \quad (M+1)\theta_{k,M} - \psi(\theta_{k,M}) = \left(k + \frac{1}{2}\right)\pi + O(\sigma^M).$$

For consecutive such zero angles in  $\mathcal{J}$ ,

$$(5.4) \quad \theta_{k+1,M} - \theta_{k,M} = \frac{\pi}{M+1} + \frac{\pi\psi'(\theta_{k,M})}{(M+1)^2} + O_{Q,r,B,\mathcal{J}}(M^{-3}).$$

In particular the zeros have local strong-clock spacing after unfolding by the Forgács–Tran parameter  $\theta$ .

*Proof.* Choose a compact interval  $\mathcal{J}_0$  in the same zero-free component with  $\mathcal{J} \subseteq \text{int } \mathcal{J}_0$ , extend  $\psi$  continuously to  $\mathcal{J}_0$ , and write

$$G_M(\theta) = \tau(\theta)^{M+1} F_M(z(\theta)), \quad \Phi_M(\theta) = (M+1)\theta - \psi(\theta).$$

On  $\mathcal{J}_0$ , the compact-interior contour expansion gives

$$G_M(\theta) = 2|W(\theta)| \cos \Phi_M(\theta) + R_M(\theta).$$

By the compact-uniform separation of Theorem 3.1, a finite cover of  $\mathcal{J}_0$  by parameter pieces carries fixed circles, each enclosing the principal pair alone and each keeping  $D$  away from zero for all  $z$  in a complex neighborhood of the corresponding piece of  $z(\mathcal{J}_0)$ . Over such a neighborhood Lemma 3.4 identifies  $R_M$  with the normalized contour remainder  $\tau^{M+1} E_M$ , bounds it by a constant times  $\sigma^M$  for some  $\sigma < 1$ , and makes  $E_M$  holomorphic in  $z$ ; Cauchy's estimate on a fixed disk then bounds  $\partial_z E_M$  by the same exponential. Since  $z$  and  $\tau$  are real-analytic on  $\mathcal{J}_0$  and differentiating  $\tau^{M+1}$  contributes at most one factor of  $M$ , this gives

$$(5.5) \quad \|R_M\|_{C^1(\mathcal{J}_0)} = O(M\sigma^M).$$

Since  $|W|$  is bounded away from zero on  $\mathcal{J}_0$  and  $\psi'$  is bounded by (3.19),

$$\Phi'_M = M+1 - \psi' = M + O(1) > 0,$$

so  $\Phi_M$  is a diffeomorphism of  $\mathcal{J}_0$  onto its image. Take  $u = \Phi_M(\theta)$  as coordinate and read  $G_M/(2|W|)$  as a function of  $u$ . Then

$$\frac{G_M}{2|W|} = \cos u + \varepsilon_M(u), \quad \|\varepsilon_M\|_{C^1} = O(\sigma^M),$$

because  $du/d\theta \asymp M$ , so the factor  $M$  in (5.5) is exactly absorbed by the change of variable. Fix  $\delta \in (0, \pi/4)$ . On  $|u - (k + \frac{1}{2})\pi| \leq \delta$  the derivative of  $\cos u + \varepsilon_M$  is bounded away from zero, and the function changes sign across the interval; outside those neighborhoods  $|\cos u| \geq \sin \delta$  exceeds  $\|\varepsilon_M\|_\infty$  for large  $M$ . Hence every such neighborhood contained in  $\Phi_M(\mathcal{J}_0)$  carries exactly one zero, and that zero is simple. Every zero of  $G_M$  whose angle lies in  $\mathcal{J}$  belongs to one of them, and there is no other zero on  $\mathcal{J}$ ; since  $\mathcal{J}$  is an interval and  $\Phi_M$  is increasing, the corresponding indices form a consecutive block. This proves (5.3). Finally  $z'(\theta) = g'(\gamma(\theta))\gamma'(\theta) \neq 0$  by (3.13) and Lemma 3.2, and  $\tau > 0$ ; differentiating  $G_M = \tau^{M+1} F_M \circ z$  at a zero of  $F_M$  therefore shows that simplicity in  $u$  or  $\theta$  is simplicity of the corresponding zero of the polynomial  $F_M$ .

For consecutive zero angles put  $\Delta_M = \theta_{k+1,M} - \theta_{k,M}$ . The phase quantization gives

$$\pi + O(\sigma^M) = (M+1)\Delta_M - (\psi(\theta_{k+1,M}) - \psi(\theta_{k,M})).$$

Since  $\Phi'_M \asymp M$ , first  $\Delta_M = O(M^{-1})$ . On  $\mathcal{J}_0$  the amplitude  $W$  is real-analytic and nonvanishing, so  $\psi$  is real-analytic and  $\psi''$  is bounded there. Taylor expansion of  $\psi$  at  $\theta_{k,M}$  then gives

$$\pi + O(\sigma^M) = (M+1 - \psi'(\theta_{k,M}))\Delta_M + O(M^{-2}).$$

Dividing by  $M+1 - \psi'(\theta_{k,M})$  and expanding its reciprocal proves (5.4); the exponentially small term is  $o(M^{-3})$ .  $\square$

For a fixed proper numerator  $N$ , apply Proposition 5.3 with  $B = B_N$  and  $M = m - \lambda_N$ , and write  $W_N, \psi_N$  for the resulting amplitude and phase. By (3.14),

$$W_N(\theta) = \frac{B_N(\gamma(\theta))}{\gamma(\theta)^r g'(\gamma(\theta))},$$

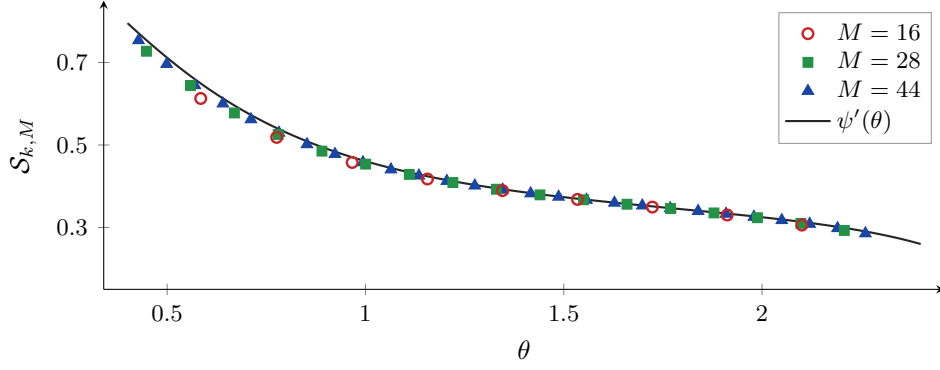


FIGURE 2. Local strong-clock collapse, for the denominator of Figure 1 and the weight  $B(t) = 1+t$ . Markers show  $\mathcal{S}_{k,M} = (M+1)^2(\theta_{k+1,M} - \theta_{k,M} - \pi/(M+1))/\pi$  at three indices, over the consecutive bulk zero angles with  $\theta_{k,M}$  and  $\theta_{k+1,M}$  both in the compact subarc  $\mathcal{J} = [0.400, 2.40]$ , on which the principal residue amplitude does not vanish. The solid curve is  $\psi'$ , computed from that amplitude alone and not fitted to the spacings. Proposition 5.3 gives  $\mathcal{S}_{k,M} = \psi'(\theta_{k,M}) + O(M^{-1})$ , and the markers tighten onto the curve as  $M$  grows. Before this rescaling the correction is under 4% of the clock  $\pi/(M+1)$  itself.

so  $\psi'_N = \Im(W'_N/W_N)$  is the sum of a denominator-only term and

$$(5.6) \quad \Im\left(\frac{B'_N(\gamma(\theta))}{B_N(\gamma(\theta))} \gamma'(\theta)\right).$$

In the shifted index the leading clock  $\pi/(M+1)$  is therefore denominator-universal, the numerator entering it only through the reindexing  $\lambda_N$ , while the canonical weight first reaches the local spacing at order  $M^{-2}$ , through the logarithmic derivative (5.6). Figure 2 shows this collapse at three indices for the fixed weight  $B(t) = 1+t$ .

**5.3. Isolated exceptional attractors.** The same residue amplitude has a one-branch manifestation. On the Forgács–Tran interval two conjugate roots are co-dominant and their interference produces the zero bulk. If instead one denominator root is strictly dominant, a zero of its residue amplitude attracts a finite packet of coefficient zeros. The result is entirely local: unlike the bounded-defect theorem, it does not require positive-rootedness of  $Q$ . That such an amplitude zero is a limit of zeros is case (i) of the Beraha–Kahane–Weiss theorem [2], restated as [19, Thm. 1.5(a)]. Exponentially rapid convergence to simple isolated limiting points is familiar in transfer-matrix examples [18], while Goh, He, and Ricci obtained the matching count for the Tribonacci pencil [14, Lems. 8–9]. The proposition below gives the exact packet multiplicity  $\nu$  and the spectral-gap rate  $O(\sigma^{M/\nu})$ ; Remark 5.5 identifies  $\nu$  with local numerator–denominator intersection multiplicity.

**Proposition 5.4.** *Let  $r \geq 1$ , let  $Q \in \mathbb{R}[t]$  satisfy  $Q(0) \neq 0$ , and let  $B \in \mathbb{R}[t] \setminus \{0\}$ . Define*

$$D(t, z) = Q(t) + zt^r, \quad F_M(z) = [t^M] \frac{B(t)}{D(t, z)}.$$

*Let  $t_0 \in \mathbb{C}^\times$  be a zero of  $B$  of multiplicity  $\nu \geq 1$ , and set*

$$z_0 = -\frac{Q(t_0)}{t_0^r}.$$

*Assume that  $t_0$  is a simple zero of  $D(\cdot, z_0)$  and is its unique minimum-modulus finite zero:*

$$(5.7) \quad |t_0| < |t| \quad \text{for every other finite zero } t \text{ of } D(t, z_0).$$

*Define the local spectral ratio*

$$(5.8) \quad \chi_0 = \max \left( \left\{ \left| \frac{t_0}{t} \right| : D(t, z_0) = 0, t \neq t_0 \right\} \cup \{0\} \right) < 1.$$

Then there is a neighborhood  $U$  of  $z_0$  such that, for every sufficiently large  $M$ ,  $F_M$  has exactly  $\nu$  zeros in  $U$ , counted with multiplicity. Moreover, for every  $\sigma \in (\chi_0, 1)$ , all of these zeros satisfy

$$(5.9) \quad |z - z_0| = O(\sigma^{M/\nu}).$$

In particular, if  $\nu = 1$ , then for all sufficiently large  $M$  there is a unique simple zero  $z_M$  of  $F_M$  near  $z_0$ , and  $z_M = z_0 + O(\sigma^M)$  for every  $\sigma \in (\chi_0, 1)$ .

*Proof.* Since  $t_0$  is a simple denominator zero, the implicit-function theorem gives a holomorphic root branch  $t = t(z)$  near  $z_0$  with  $t(z_0) = t_0$  and  $D(t(z), z) = 0$ . Moreover,

$$(5.10) \quad t'(z_0) = -\frac{\partial_z D(t_0, z_0)}{\partial_t D(t_0, z_0)} = -\frac{t_0^r}{\partial_t D(t_0, z_0)} \neq 0.$$

Put

$$\mathcal{A}(z) = -\frac{B(t(z))}{\partial_t D(t(z), z)}.$$

Because  $B$  has a zero of order  $\nu$  at  $t_0$  and  $t'(z_0) \neq 0$ ,

$$(5.11) \quad \mathcal{A}(z) = (z - z_0)^\nu \mathcal{V}(z), \quad \mathcal{V}(z_0) \neq 0.$$

Choose once an  $\eta_0 \in (\chi_0, 1)$ . By (5.7), a radius  $R_0$  can be chosen with

$$|t_0|/\eta_0 < R_0 < |t| \quad \text{for every other finite zero } t \text{ of } D(t, z_0),$$

where the right-hand restriction is vacuous if there is no other finite zero. Take  $\Gamma_0 = \{|t| = R_0\}$ . After shrinking the  $z$ -neighborhood,  $t(z)$  is the only denominator zero inside  $\Gamma_0$  and  $|t(z)|/R_0 \leq \eta_0$ . Hence Lemma 3.4 gives

$$(5.12) \quad t(z)^{M+1} F_M(z) = \mathcal{A}(z) + O(\eta_0^M)$$

uniformly there. Choose a small disk  $U$  centered at  $z_0$ , contained in that neighborhood, on whose boundary  $\mathcal{A}$  does not vanish and in which  $z_0$  is its only zero. Rouché's theorem applied to (5.12) shows that  $F_M$  has exactly  $\nu$  zeros in  $U$  for every sufficiently large  $M$ .

Now fix  $\sigma \in (\chi_0, 1)$  and choose  $\eta \in (\chi_0, \sigma)$ . Repeating the same origin-centered separating-circle construction with  $\eta$  gives, on some neighborhood contained in  $U$ ,

$$t(z)^{M+1} F_M(z) = \mathcal{A}(z) + O(\eta^M).$$

On the circle  $|z - z_0| = C_* \eta^{M/\nu}$ , (5.11) gives  $|\mathcal{A}(z)| \geq c C_*^\nu \eta^M$  for all sufficiently large  $M$ , while the remainder is at most  $C \eta^M$ . Choosing  $C_*$  sufficiently large and applying Rouché again shows that the shrinking disk contains  $\nu$  zeros. Since the fixed disk  $U$  contains exactly  $\nu$ , these are the same zeros, and

$$|z - z_0| = O(\eta^{M/\nu}) = O(\sigma^{M/\nu}).$$

When  $\nu = 1$ , the single zero has multiplicity one. □

**Remark 5.5.** The packet multiplicity is intrinsic before reduction. For the canonical Laurent restriction of Lemma 2.2,

$$N(t, g(t)) = L_N(t) = t^{\lambda_N} B_N(t).$$

At a nonzero simple denominator root  $t_0$ , multiplication by the monomial  $t^{\lambda_N}$  does not change vanishing order, while

$$g'(t_0) = -\frac{\partial_t D(t_0, z_0)}{t_0^r} \neq 0.$$

Thus the denominator curve is smooth and

$$(5.13) \quad \nu = \text{ord}_{t_0} B_N = \text{ord}_{t_0} N(t, g(t)) = I_{(t_0, z_0)}(N, D),$$

the local intersection multiplicity of the numerator and denominator curves. The proposition therefore says that a uniquely dominant cancellation of intersection multiplicity  $\nu$  attracts exactly  $\nu$  coefficient zeros. If  $z_0 \notin [0, \infty)$ , these are persistent exceptional zeros; the closed ray is needed, since a packet attracted to  $z_0 = 0$  may approach it from inside  $(0, \infty)$ .

The exceptional pair in Figure 1(B) is an exact instance of Proposition 5.4.

**Corollary 5.6.** For  $Q(t) = (1-t)(1-t/2)(1-t/4)$ ,  $r = 1$ , and  $N(t, z) = 1 + z + z^2 + t(2 - z)$ , let  $N/(Q + zt) = \sum_{m \geq 0} P_m(z)t^m$ . Then

$$[z^{m+2}]P_m = (-1)^m, \quad \deg P_m = m + 2 \quad (m \geq 0).$$

There is a nonreal algebraic point  $z_* = -0.5655268358 \dots + 1.3674915753 \dots i$  such that, for all sufficiently large  $m$ ,  $P_m$  has a unique simple zero  $z_m$  near  $z_*$  with  $z_m = z_* + O(3^{-m})$ , and the conjugate zero converges at the same rate to  $\bar{z}_*$ .

*Proof.* Coefficient comparison gives

$$P_0 = z^2 + z + 1, \quad P_1 = -z^3 + \frac{3}{4}z^2 - \frac{1}{4}z + \frac{15}{4}, \quad P_m = -\left(z - \frac{7}{4}\right)P_{m-1} - \frac{7}{8}P_{m-2} + \frac{1}{8}P_{m-3},$$

with  $P_{-1} = 0$ . Thus  $-zP_{m-1}$  is inductively the unique term of degree  $m + 2$ , proving the first assertion.

Direct substitution in the canonical Laurent restriction gives

$$N(t, g(t)) = t^{-2}B(t), \quad 64B(t) = t^6 - 22t^5 + 141t^4 - 252t^3 + 548t^2 - 288t + 64.$$

Thus  $\lambda_N = -2$ , and Lemma 2.2 gives  $P_m = F_{m+2}$  for all sufficiently large  $m$ . Write  $64B = \mathcal{Q} + \mathcal{H}$ , with  $\mathcal{Q} = 548t^2 - 288t + 64$  and  $\mathcal{H} = t^6 - 22t^5 + 141t^4 - 252t^3$ . On  $|t| = 1/2$ ,

$$|\mathcal{H}(t)| \leq \frac{2625}{64} < \sqrt{\frac{298424}{137}} \leq |\mathcal{Q}(t)|,$$

the last inequality following by minimizing  $|\mathcal{Q}(\frac{1}{2}e^{i\theta})|^2 = 35072 \cos^2 \theta - 57888 \cos \theta + 26065$ . Since the two zeros  $(36 \pm 8i\sqrt{14})/137$  of  $\mathcal{Q}$  have modulus  $4/\sqrt{137} < 1/2$ , Rouché gives exactly two zeros of  $B$  in  $|t| < 1/2$ . They are nonreal:  $64B(t) > 0$  termwise on  $[-1/2, 0]$ , while on  $[0, 1/2]$

$$\mathcal{Q}(t) = 548 \left(t - \frac{36}{137}\right)^2 + \frac{3584}{137} \geq \frac{3584}{137}, \quad \mathcal{H}(t) \geq \mathcal{H}(1/2) = -\frac{1495}{64},$$

the second inequality because the cubic factor in  $\mathcal{H}'(t) = 2t^2(3t^3 - 55t^2 + 282t - 378)$  has derivative at least  $917/4$  on  $[0, 1/2]$ , hence increases there and is bounded above by its value  $-2003/8$  at  $1/2$ ; so  $\mathcal{H}' \leq 0$  and  $\mathcal{H}$  decreases. Hence  $64B \geq 24561/8768 > 0$ . Thus the inner zeros form a simple conjugate pair. Let  $t_*$  be the upper-half-plane one and set  $z_* = -Q(t_*)/t_*$ .

If  $u, v$  are the other denominator zeros at  $z_*$ , then  $-8(Q(t) + z_*t)$  is  $t^3 - 7t^2 + (14 - 8z_*)t - 8$ , and Vieta gives

$$u + v = 7 - t_*, \quad uv = \frac{8}{t_*}.$$

Were  $|u| \leq 3/2$ , then  $|v| > 32/3$  and hence  $|u+v| > 55/6$ , contradicting  $|7-t_*| < 15/2$ ; the same argument applies to  $v$ . Thus  $|u|, |v| > 3/2$ , so  $t_*$  is the unique minimum-modulus denominator zero and the local spectral ratio of (5.8) is less than  $\frac{1/2}{3/2} = \frac{1}{3}$ . Moreover,  $z_* \notin \mathbb{R}$ , since otherwise  $\bar{t}_*$  would be another denominator zero of modulus  $|t_*|$ . Applying Proposition 5.4 with  $\nu = 1$ ,  $\sigma = 1/3$ , and  $P_m = F_{m+2}$  gives  $z_m - z_* = O(3^{-m})$ , and conjugation gives the second zero.

Finally Remark 5.5 gives  $N(t_*, z_*) = 0$ . Since  $N(t, 2) = 7$ , solving  $N(t, z) = 0$  for  $t$  and substituting in the denominator gives

$$Q\left(\frac{z^2 + z + 1}{z - 2}\right) + z \frac{z^2 + z + 1}{z - 2} = -\frac{z^6 - 12z^5 + 44z^4 - 44z^3 + 96z^2 - 104z + 135}{8(z - 2)^3}.$$

Thus  $z_*$  is algebraic; the displayed decimal merely selects the relevant conjugate root.  $\square$

## 6. CONCLUDING REMARKS

Taken together, the results give a single spectral description of what fixed initial conditions change. On the denominator curve  $D(t, z) = 0$ , the numerator restricts canonically to

$$L_N(t) = N(t, g(t)) = t^{\lambda_N} B_N(t).$$

The monomial factor  $t^{\lambda_N}$  produces only an index and degree shift. The remaining weight  $B_N$  acts at two scales. Its degree controls the eventual global discrepancy, while its restriction to

the dominant spectral locus controls the local fine structure. On a zero-free co-dominant arc its logarithmic derivative supplies the numerator-dependent term in the first clock correction (5.6); zeros of  $B_N$  suppress the principal amplitude and, on uniquely dominant branches, produce cancellation packets whose size is, by Remark 5.5, the local numerator–denominator intersection multiplicity. Thus, apart from the index and degree shift carried by  $\lambda_N$ , the denominator determines the bulk zero law: Theorem 1.1 and Proposition 5.1 show that the numerator contributes at most a bounded global defect and does not alter the limiting zero distribution.

The contour-separated expansion is the common perturbative mechanism. Only fibers of  $g(t) = -Q(t)/t^r$  whose moduli can compete need to be resolved individually; every other pole is exponentially negligible. For a co-dominant conjugate pair the resulting normal form is oscillatory: globally its sign changes survive amplitude zeros and endpoint clusters to give the bounded angular discrepancy (4.17), while on a zero-free compact subarc the same normal form sharpens to the phase quantization and strong-clock law of Proposition 5.3. For a uniquely dominant branch the normal form is instead a single analytic amplitude plus an exponentially small remainder, and its zeros produce the exceptional packets of Proposition 5.4. In this sense co-dominance quantizes the zero bulk, whereas unique dominance localizes cancellations. The positive-rootedness of  $Q$  supplies the global Forgács–Tran dominant-pair geometry; the Laurent reduction, contour expansion, local phase quantization, and isolated-cancellation mechanism themselves are more general.

The order  $O(1)$  of the defect is optimal. The factorization  $N(t, z) = R(z)$  gives  $P_m(z) = R(z)H_m(z)$ , so an arbitrary fixed collection of exceptional zeros can persist at every index. Such zeros need not come from a fixed factor: Proposition 5.4 and Corollary 5.6 exhibit a second mechanism in which exceptional zeros converge exponentially to algebraic cancellation points.

**Question 6.1.** Do there exist constants  $C_0(Q, r)$  and  $C_1(Q, r)$ , depending only on the admissible denominator, such that for every  $B \in \mathbb{R}[t]$  with  $B(0) \neq 0$  and every  $M \geq 0$  with  $F_M \neq 0$ ,

$$\#\{\text{zeros of } F_M \text{ outside } I_{Q,r}\} \leq C_0(Q, r) + C_1(Q, r) \deg B,$$

counted with multiplicity?

The issue is genuinely all-index rather than one of uniform onset: Remark 2.5 gives degree-one weights with degree drops at arbitrarily large indices, so a positive answer cannot follow merely by making the onset in Proposition 4.4 uniform in  $\deg B$ . The Laurent reduction introduces no analogous obstruction, since Lemma 2.2 bounds both its transient and  $\deg B_N$  in terms of the numerator bidegrees; a positive answer would therefore bound the original sequence in terms of those bidegrees and the fixed denominator data. In the quadratic case Remark 3.3 gives the stronger reduced bound  $\deg B + 2$  at every index, and for the pencil  $1 + s + t + zst$  Luong and Tran obtain the explicit numerator-degree bound  $\deg_s \tilde{N} + \deg_t \tilde{N} + 2 \deg_z \tilde{N}$  [16].

Whether the asymptotic obstructions found here are the only ones is open.

**Question 6.2.** Can eventual positive real-rootedness be characterized entirely from the canonical weight  $B_N$  on the dominant spectral locus? More precisely, are the only asymptotic obstructions

- (1) zeros of  $B_N$  on a simple uniquely dominant branch over  $z_0 \notin [0, \infty)$ , which produce the exceptional packets of Proposition 5.4; and
- (2) zeros of the principal amplitude  $W$  on the co-dominant Forgács–Tran arc, where the leading two-branch oscillation is suppressed, together with a local sign or interlacing condition there?

If so, is all-index positive real-rootedness characterized by these spectral conditions together with the finitely many pre-reduction coefficient polynomials?

This is a numerator-sensitive refinement of the operator problem posed by Forgács and Tran in their Problem 15 [11]. Necessary and sufficient conditions of this kind are known for a linear combination of Chebyshev polynomials with coefficients the powers of  $az + b$  [22].



Finally, the proof leaves a denominator-side question that is independent of the particular numerator.

**Question 6.3.** Which real denominator pencils  $D(t, z)$  admit a real dominant-pair locus  $\mathcal{L}$  such that, for every fixed proper polynomial numerator, all but  $O(1)$  coefficient zeros lie on  $\mathcal{L}$ , uniformly in the coefficient index? Do intrinsic dominant-pair and endpoint-separation conditions suffice?

The first test case is the linear-type denominator  $D(t, z) = Q_0(t) + zt^r Q_1(t)$ , with nonconstant  $Q_1$ , studied under separation hypotheses by Forgács and Tran [13]. Here the canonical restriction is rational rather than polynomial, with fixed poles at the zeros of  $Q_1$ . Does the bounded-defect theorem survive whenever those poles remain outside the dominant separating contours, and what local zero law appears at the first parameter where such a pole meets a dominant branch or endpoint cluster?

## DECLARATIONS

**Funding.** This research received no external funding.

**Competing interests.** The author has no relevant interests to disclose.

**Data availability.** No datasets were generated or analyzed.

**Computational verification.** The proofs do not depend on numerical computation. Scripts used for verification and figures are available at <https://github.com/johnnyshields/papers>.

**Generative AI use.** During the preparation of this work, the author used OpenAI’s ChatGPT (GPT-5.6 Sol) and Anthropic’s Claude (Opus 5, Opus 4.8) to assist with mathematical research, manuscript drafting, and the development of auxiliary verification code. After using these tools, the author reviewed and edited as needed. The author takes full responsibility for the content of the work.

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